

Purely Analytic Supplement for A Dirichlet Pólya Inequality on Three-Dimensional Spherical Shells

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Abstract

This support volume contains the detailed hand proofs and exact-rational tables used to remove every executable predicate from the logical dependency chain of the spherical-shell theorem. Parts I–VI replace the 611-row exact ledger; Parts VII–IX replace the compact interval certificate; Parts X–XIII replace the aggregate base certificate; and Part XIV assembles the final residual closure. Programs and interval certificates remain available only as optional regression checks and are not premises of this analytic proof.

Reading convention. The volume is modular so that each bottleneck lemma can be reviewed and hashed independently. Every included part is generated from its corresponding LaTeX source in the `manuscript/analytic` directory. The repeated title pages deliberately mark the trust boundary between exhibits.

Global census of the exact ledger

The canonical ledger has 553 core rows and 58 supplemental rows. The following inclusive ranges give a one-to-one allocation; the part numbers refer to this volume. The family codes denote, in order, rational- π checks (PI), thresholds (TH), strict-floor conventions (SF), zero specializations (Z), lower inventories (LI), checkpoint inventories (KI), cap payments (CP), localizations (LOC), the upper owner (U), optical screens (OPT), exact subtraction (SUB), and supplemental seam/payment rows (SUP).

canonical family	row range	unique owner
PI	PI1–PI4	II
TH	TH1–TH7, TH31–TH34 / TH8–TH30	I / II
SF	SF1–SF35	II
Z	Z1–Z48	I
LI	LI1–LI8, LI10–LI44 / LI9	I / II
KI	KI1–KI117	II
CP	CP1–CP43, CP56, CP60–CP77 / remaining rows	II / III
LOC	LOC1–LOC47	III
U	U1–U8	V
OPT	OPT1–OPT14 / OPT15–OPT50	V / II
SUB	SUB1–SUB68	VI
SUP	SUP1–SUP57 / SUP58	IV / I

Here “remaining rows” in CP means CP44–CP55, CP57–CP59, and CP78–CP98. Thus the six disjoint owner totals are

$$103 + 278 + 83 + 57 + 22 + 68 = 611.$$

In particular, the ten lower- d identities SUP36–SUP45 are assigned to Part IV although Part II also proves their general odd-sum identity; SUP58 is assigned to Part I; and OPT15–OPT50 are assigned to Part II although Part V reuses their consequences. These are proof redundancies, not row overlaps. The displayed ranges exhaust every canonical row and have empty pairwise intersections.

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Hand-visible analytic derivations for 103 finite-ledger predicates

1 Scope and logical convention

This note exposes, in hand-checkable form, the predicates not assigned to the other five ledger packets. They are precisely

11 threshold predicates + 48 zero predicates + 43 lower-closure predicates + 1 cap-74 predicate = 103.

The omitted lower-closure predicate is the single six-cap tuple identity; it belongs to the cap-inventory packet. Every inequality below is proved from a displayed analytic inequality, integer arithmetic, rational cross multiplication, or positive-side squaring. An executable ledger may check the same arithmetic, but no executable output is used as a premise.

We use

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \quad C_0 = 1 + \frac{8\sqrt{2}}{15}, \quad \rho_* = \frac{\omega_0}{2C_0}, \quad \rho_0 = \frac{1}{\sqrt{337}}, \quad \sigma = \frac{3\omega_0}{4}, \quad \rho_c = \frac{1}{1+2\pi}, \quad d = \frac{\sqrt{337}}{2}.$$

The elementary enclosure

$$\frac{333}{106} < \pi < \frac{355}{113} < \frac{22}{7} \tag{1}$$

is analytic: the lower and middle bounds follow from alternating truncations of Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and the last one is immediate by cross multiplication. For reference, 355/113 exceeds the upper Machin truncation by the positive rational numerator 45167474711 over denominator 189820334689453125. The classical integral

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi$$

also gives the final upper bound directly.

2 The eleven threshold predicates

First we obtain a rational two-sided enclosure for ω_0 . The exact integer residuals

$$3 \, 1101^2 113^2 - 607^2 355^2 = 1998482 > 0, \quad 25 \, 333^2 - 243 \, 106^2 = 41877 > 0$$

and (1) imply

$$\sqrt{3} > \frac{607 \, 355}{1101 \, 113} > \frac{607\pi}{1101}, \quad 9\sqrt{3} < 5\pi.$$

Consequently

$$\frac{40}{367} < \omega_0 < \frac{1}{9}. \tag{2}$$

Indeed, $607/2202 - 1/6 = 40/367$, while $5/18 - 1/6 = 1/9$.

Since $2 \cdot 32^2 - 45^2 = 23 > 0$, one has $\sqrt{2} > 45/32$, hence $C_0 > 7/4$ and $\rho_* < 2/63$. The following chain consists entirely of strict rational comparisons, with the indicated radical comparisons justified by squaring:

$$\rho_* < \frac{2}{63} < \frac{1}{\sqrt{337}} < \frac{1}{18} < \frac{3}{40} < \frac{3\omega_0}{4} < \frac{1}{12} < \frac{7}{51} < \frac{1}{1+2\pi} < \frac{1}{7} < \frac{39}{50} < \frac{7}{8}.$$

Here

$$4 \cdot 337 < 63^2, qquad 18^2 < 337, qquad 367 < 400, \quad 51 < 84, qquad 367 < 420,$$

and $7/51 < \rho_c < 1/7$ follows respectively from $\pi < 22/7$ and $\pi > 3$. This proves the five ratio-order rows. It also proves

$$\omega_0 < \frac{1}{9} < \frac{7}{51} < \rho_c.$$

Moreover

$$\omega_0 d > \frac{20\sqrt{337}}{367} > 1, \quad 400 \cdot 337 - 367^2 = 111 > 0.$$

The four terminal comparisons follow from the same bounds:

$$\begin{aligned} \sqrt{114} &< 18 < \frac{2}{\omega_0}, & \frac{2}{\omega_0} &< \frac{367}{20}, \\ \rho_c &< \frac{1}{7} < \frac{60}{367} < \frac{3\omega_0}{2}, & \omega_0 \sqrt{114} &> \frac{40\sqrt{114}}{367} > 1. \end{aligned}$$

The last positive-side squaring has residual

$$1600 \cdot 114 - 367^2 = 47711 > 0.$$

Thus the one-to-one threshold registry is

ID	source row	proved predicate
T1–T5	1–5	$\rho_* < \rho_0 < \sigma < \rho_c < 39/50 < 7/8$, one adjacent pair per row
T6	6	$\omega_0 < \rho_c$
T7	7	$\omega_0 d > 1$
T8	31	$\sqrt{114} < 2/\omega_0$
T9	32	$2/\omega_0 < 367/20$
T10	33	$\rho_c < 3\omega_0/2$
T11	34	$\omega_0 \sqrt{114} > 1$

3 The forty-eight zero specializations

3.1 A recurrence and elementary tangent bounds

Put

$$j_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+1/2}(x), \quad P_\ell(x) = x^{\ell+1} j_\ell(x).$$

Then

$$P_{\ell+1} = (2\ell+1)P_\ell - x^2 P_{\ell-1}, \quad P_0 = \sin x, \quad quad P_1 = \sin x - x \cos x. \quad (3)$$

Write $P_\ell = A_\ell(t) \sin x + x C_\ell(t) \cos x$, $t = x^2$. The recurrence gives

$$\begin{aligned} A_2 &= 3 - t, & C_2 &= -3, \\ A_3 &= 15 - 6t, & C_3 &= t - 15, \\ A_5 &= 945 - 420t + 15t^2, & C_5 &= -945 + 105t - t^2, \\ A_6 &= 10395 - 4725t + 210t^2 - t^3, & C_6 &= -10395 + 1260t - 21t^2, \\ A_7 &= 135135 - 62370t + 3150t^2 - 28t^3, & C_7 &= -135135 + 17325t - 378t^2 + t^3. \end{aligned}$$

For $0 < s < \pi/2$, differentiation and integration give

$$\tan s > s, \text{quad} \tan s > s + \frac{s^3}{3}. \quad (4)$$

For $0 < s < 1$, $\sin s < s$ and $\cos s > 1 - s^2/2$ give

$$\tan s < \frac{s}{1 - s^2/2}. \quad (5)$$

3.2 Sixteen half-period locations

For each internal face, the first displayed margin below is $w - (h_L/2)(22/7)$; the second is $(h_R/2)(333/106) - w$. Positivity therefore proves both strict half-period location rows without a decimal approximation.

(ℓ, n)	w	(h_L, h_R)	left margin	right margin
(1, 2)	77/10	(4, 5)	99/70	163/1060
(2, 2)	9	(5, 6)	8/7	45/106
(3, 2)	103/10	(6, 7)	61/70	737/1060
(5, 2)	129/10	(8, 9)	23/70	1311/1060
(1, 3)	21/2	(6, 7)	15/14	105/212
(2, 3)	61/5	(7, 8)	6/5	97/265
(6, 1)	10	(6, 7)	4/7	211/212
(7, 1)	23/2	(7, 8)	1/2	113/106

3.3 Sixteen half-period signs

At an integral or half-integral multiple of π , exactly one of $\sin x, \cos x$ vanishes. Direct substitution in the displayed polynomials gives the following complete sign table.

(ℓ, n)	h_L	left sign	(ℓ, n)	h_R	right sign
(1, 2)	4	—	(1, 2)	5	+
(2, 2)	5	—	(2, 2)	6	+
(3, 2)	6	—	(3, 2)	7	+
(5, 2)	8	—	(5, 2)	9	+
(1, 3)	6	+	(1, 3)	7	—
(2, 3)	7	+	(2, 3)	8	—
(6, 1)	6	+	(6, 1)	7	—
(7, 1)	7	+	(7, 1)	8	—

Here are explicit polynomial witnesses for the only signs not immediate from $A_1 = 1, C_1 = -1, A_2 = 3 - t, C_2 = -3, A_3 = 15 - 6t, C_3 = t - 15$. From (1), at $x = 4\pi$ one has $144 < t < 159$, at $x = 9\pi/2$ one

has $t > 182$, and at $x = 7\pi/2$ one has $110 < t < 121$. Monotonicity on these rational intervals and direct evaluation give

$$C_5(t) < C_5(144) = -6561, \text{quad} A_5(t) > A_5(182) = 421365,$$

$$C_6(t) < C_6(81) = -46116, \text{quad} A_6(t) > A_6(110) = 700645,$$

$$A_7(t) < A_7(110) = -5878565, \text{quad} C_7(t) < C_7(144) = -2492559.$$

For example, $A'_6 = -4725 + 420t - 3t^2$ is positive on $[110, 121]$, while A'_7 is negative there; the other asserted monotonicities follow at once from their displayed derivatives. Combining these signs with the quadrant signs of sine and cosine proves all sixteen entries.

3.4 Eight rational-wall signs

Substitution in (3) reduces each rational-wall sign to one of the following exact tangent comparisons. The shifted angle is always in $(0, \pi/2)$, as already proved by the location table.

(ℓ, n)	wall	hand certificate	$\text{sgn } j_\ell(w)$
(1, 2)	77/10	For $u = 5\pi/2 - w$, $\tan u > 3/20 > 10/77 = 1/w$; hence $\tan(w - 2\pi) = \cot u < w$.	−
(2, 2)	9	For $u = 3\pi - w$, $\tan u > 21/50 > 9/26 = 3w/(w^2 - 3)$.	−
(3, 2)	103/10	For $u = 7\pi/2 - w$, $\tan u > 69/100 > 621540/938227 = (6w^2 - 15)/(w(w^2 - 15))$.	−
(5, 2)	129/10	For $y = w - 4\pi$, $0 < y < 17/50$, so $\tan y < 1700/4711 < 177800829/427700150 = -wC_5/A_5$.	−
(1, 3)	21/2	For $u = 7\pi/2 - w$, $\tan u > 49/100 > 2/21 = 1/w$, hence $\tan(w - 3\pi) < w$.	+
(2, 3)	61/5	For $u = 4\pi - w$, $\tan u > 9/25 > 915/3646 = 3w/(w^2 - 3)$.	+
(6, 1)	10	For $y = w - 3\pi$, $4/7 < y < 29/50$, so $\tan y < 2900/4159 < 188790/127579 = -wC_6/A_6$.	+
(7, 1)	23/2	For $u = 4\pi - w$, $\tan u > u + u^3/3 > 546377/375000 > 3153151489/2276518664 = wC_7/A_7$.	+

For complete cross-multiplication visibility, the four less immediate positive gaps in that table are

$$\begin{aligned} \frac{69}{100} - \frac{621540}{938227} &= \frac{2583663}{93822700}, \\ \frac{177800829}{427700150} - \frac{1700}{4711} &= \frac{110529450419}{2014895406650}, \\ \frac{188790}{127579} - \frac{2900}{4159} &= \frac{415198510}{530601061}, \\ \frac{546377}{375000} - \frac{3153151489}{2276518664} &= \frac{3837851856583}{53355906187500}. \end{aligned}$$

The four small gaps are respectively $31/1540$, $24/325$, $829/2100$, $9939/91150$.

3.5 Six Lorch rows and two wall-order rows

For $\nu = \ell + 1/2$, Lorch's strict analytic first-zero estimate is

$$j_{\nu,1}^2 > \frac{24(\nu + 1)^2}{1 - 2\nu + \sqrt{(2\nu + 3)(2\nu + 11)}} - 2(\nu^2 - 1) =: R_\ell.$$

After rationalization,

$$R_\ell = \frac{3(2\ell + 3)\sqrt{(\ell + 2)(\ell + 6)} - 2\ell^2 + \ell + 6}{4}.$$

All sides being squared are positive, and the following integer residuals prove the three required strict specializations:

ℓ	wall c	positive squared residual proving $R_\ell > c^2$
8	71/6	$1026^2 \cdot 35 - 6067^2 = 35171$
9	64/5	$1575^2 \cdot 165 - 20059^2 = 6939644$
10	69/5	$3450^2 \cdot 3 - 5911^2 = 767579$

Since $j_\ell(x) > 0$ for sufficiently small $x > 0$, and its first positive zero is strictly to the right of c , these same three rows imply $j_\ell(c) > 0$. Thus the three executable sign rows are analytic corollaries, not independent numerical assumptions. Finally,

$$\frac{103}{10} - \frac{81}{8} = \frac{7}{40} > 0, \quad \frac{61}{5} - \frac{21}{2} = \frac{17}{10} > 0.$$

The one-to-one zero registry is as follows. Number the eight internal faces in the order of the location table. For face r , IDs $Z_{5r-4}, \dots, Z_{5r}$ are, in order: right of the left half-period, left of the right half-period, left half-period sign, right half-period sign, and rational-wall sign. This bijects IDs $Z1$ – $Z40$ with the first forty source rows. The remaining bijection is

ID	source row	predicate
Z41	41	Lorch specialization $\ell = 8, c = 71/6$
Z42	42	positive sign at $\ell = 8, c = 71/6$
Z43	43	Lorch specialization $\ell = 9, c = 64/5$
Z44	44	positive sign at $\ell = 9, c = 64/5$
Z45	45	Lorch specialization $\ell = 10, c = 69/5$
Z46	46	positive sign at $\ell = 10, c = 69/5$
Z47	47	$103/10 > 81/8$
Z48	48	$61/5 > 21/2$

4 The forty-three lower-closure predicates

4.1 Eight activation and exclusion rows

The complete arithmetic is

$$\begin{aligned}
(23/2)^2 - 114 &= 73/4 > 0, & 10^2 &= 100, \\
(129/10)^2 - 114 &= 5241/100 > 0, & 103/10 - 81/8 &= 7/40 > 0, \\
(81/8)^2 + 8 &= 7073/64, & (81/8)^2 + 18 - 114 &= 417/64 > 0, \\
(21/2)^2 + 4 - 114 &= 1/4 > 0, & 16\pi^2 &> 16 \cdot 3^2 > 114.
\end{aligned}$$

These are exactly lower IDs $L1$ – $L8$, corresponding to source rows 1–8.

4.2 Five Weyl payments

For $0 < \rho < \rho_c < 1/7$, (1) yields the uniform analytic lower bound

$$W(\rho, K) = \frac{2}{9\pi}(1 - \rho^3)K^3 > \frac{38}{539}K^3. \quad (6)$$

Using $d > 9$ and $\sqrt{7073}/8 > 21/2$, the five payments are the following exact positive reserves:

ID	source row	threshold/cap	reserve from (6)
L9	10	$K > d, 46$	$(38/539)9^3 - 46 = 2908/539$
L10	11	$K > 10, 59$	$(38/539)10^3 - 59 = 6199/539$
L11	12	$K > 81/8, 66$	$(38/539)(81/8)^3 - 66 = 990435/137984$
L12	13	$K > 21/2, 69$	$(38/539)(21/2)^3 - 69 = 555/44$
L13	14	$K > \sqrt{7073}/8, 78$	$(38/539)(21/2)^3 - 78 = 159/44$

4.3 The cap-395 cell: eighteen rows

Put $R = 367/20$, $z_\ell = \ell + 1/2$. The analytic turning estimate

$$G_K(z) < \frac{(K - z)^{3/2}}{3\sqrt{K}}$$

is increasing in K , so on $K < R$ a strict cap c_ℓ follows from

$$\frac{(R - z_\ell)^3}{9R} = \frac{(357 - 20\ell)^3}{1321200} < \left(c_\ell + \frac{3}{4}\right)^2. \quad (7)$$

The following table gives all fifteen caps and the exact positive numerator $82575(4c_\ell + 3)^2 - (357 - 20\ell)^3$ obtained by clearing the positive denominators in (7).

ℓ	0	1	2	3	4	5	6	7
c_ℓ	6	5	5	4	4	3	3	3
margin	14697882	5409422	11827162	3611502	8555642	1604782	5267322	8361062

ℓ	8	9	10	11	12	13	14
c_ℓ	2	2	1	1	1	1	0
margin	2346202	4446342	176282	1474822	2444562	3133502	286642

Thus L15–L29 correspond bijectively to source rows 16–30, with L($15 + \ell$) the ℓ -th table column. Since $G_R(z)$ is strictly decreasing in z , the $c_{14} = 0$ row removes every $\ell \geq 14$ tail; this is L30/source row 31.

The weighted cap itself, L14/source row 15, is visible from the contributions

$$6, 15, 25, 28, 36, 33, 39, 45, 34, 38, 21, 23, 25, 27, \quad \sum_{\ell=0}^{13} (2\ell + 1)c_\ell = 395.$$

Finally, on the exceptional cell $\rho K \geq 5/2$, with $\rho < 11/80$ and $1/\pi > 113/355$,

$$\begin{aligned} W(\rho, K) &> \frac{2}{9} \frac{113}{355} \left(\frac{5}{2}\right)^3 \left[\left(\frac{80}{11}\right)^3 - 1 \right] \\ &= \frac{160293325}{378004} = 395 + \frac{10981745}{378004} > 395. \end{aligned}$$

This is L31/source row 32.

4.4 Twelve floor and boundary-owner rows

The nine finite arithmetic rows are instances of one identity:

$$\frac{3}{2}(p+1) \leq 2p + \frac{1}{2} \quad (p \geq 2), \text{quad} \left(2p + \frac{1}{2}\right) - \frac{3}{2}(p+1) = \frac{p-2}{2} \geq 0. \quad (8)$$

At $p = 2$, $(3/2)3 = 9/2$; this is L32/source row 33. For $p = 2, \dots, 9$, formula (8) is respectively L33–L40/source rows 34–41.

The half-integer face is exact: $5/2 = 2 + 1/2$. Assigning it to the high side makes the interface remainder vanish and leaves the exceptional cell closed; this is L41/source row 42. At $K = 2/\omega_0$,

$$p = \lfloor \omega_0 K \rfloor = 2, \quad \rho K \leq \frac{2\rho_c}{\omega_0} < 3$$

by T10. This proves L42/source row 43. Moreover

$$\rho K - \frac{1}{2} < \frac{5}{2}, \quad m = \left\lfloor \rho K - \frac{1}{2} \right\rfloor \leq 2 \leq 2p - 1 = 3,$$

and the strict exceptional condition $K < 2/\omega_0$ fails at equality. Thus the equality face is aggregate-owned, proving L43/source row 44. More generally, if $K \geq 2/\omega_0$, then $p \geq 2$ and

$$\rho K < \frac{3}{2}(p + 1) \leq 2p + \frac{1}{2}, \quad m \leq 2p - 1,$$

so the finite instances conceal no unproved large- p case.

5 The supplemental cap–74 incompatibility

The propagated ball wall requires $z^2 > 2409/100$, while the product wall requires $z^2 < 70/3$. They are incompatible because

$$\frac{2409}{100} - \frac{70}{3} = \frac{7227 - 7000}{300} = \frac{227}{300} > 0.$$

This is supplemental ID S1 and is a hand proof, not merely a reported certificate value.

6 Census and source-visibility verdict

family	count	IDs	hand-visible status before this note
threshold remainder	11	T1–T11	printed, with a few implications compressed
zero specializations	48	Z1–Z48	tables printed; several recurrence steps compressed
lower remainder	43	L1–L43	payments printed; floor instances generated implicitly
supplemental cap–74	1	S1	printed exactly
total	103		

No predicate in this 103-row allocation is an unsupported executable-only mathematical premise. Before this note, the principal visibility gaps were the one-to-one expansion of the forty zero sign/location rows, the fact that the three Lorch-wall signs are logical corollaries of the strict first-zero bounds, and the expansion of the twelve floor/boundary rows. The displayed recurrences, rational margins, general floor identity, and exact ownership arguments close those presentation gaps.

Exact-ledger decomputerization: Packets A and B

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Abstract

This note replaces two groups of executable exact-arithmetic checks in the three-dimensional spherical-shell Pólya proof. Packet A proves once and for all the strict-bracket convention, odd-multiplicity block sums, the optical product-deficit identity, and monotonicity of propagated zero and channel cuts. Packet B gives exact hand-checkable tables for the five moving $3z$ -crossings and for the seven first-excluded checkpoint channels. It then proves, without a finite search, that the five printed checkpoint inventories are exhaustive. Every numerical comparison below is an equality of rational numbers, a cross multiplication, or a positive-side squaring. No program output, decimal approximation, or interval calculation is a premise.

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1 Scope and notation

Throughout $0 < \rho < 1$. Put

$$z = z_\rho := \frac{\pi}{1 - \rho}, \quad k_m(\rho)^2 := z_\rho^2 + m(m + 1),$$

and define the two ratio endpoints

$$\rho_0 := \frac{1}{\sqrt{337}}, \quad \rho_c := \frac{1}{1 + 2\pi}.$$

The high-checkpoint argument uses the strict proxy

$$b_{\ell,n}(z)^2 := \max\{n^2 z^2 + \ell(\ell + 1), \beta_{\ell,n}^2\}. \quad (1)$$

A channel may be counted at K only when $b_{\ell,n}(z) < K$. The constants $\beta_{\ell,n}$ are strict lower bounds for the corresponding ball zeros. The zero bounds used in the checkpoint table are displayed explicitly in Section 5.

The purpose of this note is deliberately limited. It replaces the general exact-arithmetic and checkpoint-ordering rows of the old ledger; it does not replace the zero-sign proofs, high-event payment registry, seam arithmetic, optical endpoint screens, or formal owner subtraction.

2 A rational enclosure for π

Lemma 2.1 (Hand enclosure for π). *One has*

$$\frac{333}{106} < \pi < \frac{22}{7}, \quad \frac{1}{\pi} < \frac{106}{333}. \quad (2)$$

Proof. For $0 < x \leq 1$, the alternating series for $\arctan x$ gives a strict lower bound after a negative term and a strict upper bound after a positive term. Hence

$$\arctan \frac{1}{5} > \frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} - \frac{1}{7 \cdot 5^7} = \frac{323852}{1640625}, \quad \arctan \frac{1}{239} < \frac{1}{239}.$$

Machin's identity therefore yields

$$\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239} > \frac{1231847548}{392109375},$$

and direct subtraction gives the positive margin

$$\frac{1231847548}{392109375} - \frac{333}{106} = \frac{3418213}{41563593750} > 0.$$

For the upper bound, polynomial division and elementary integration give

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi.$$

Finally, reciprocation of the positive lower bound for π gives the last inequality. Notice also that $22/7 - 333/106 = 1/742 > 0$, so the enclosure is nonempty. $\square$

3 Packet A: universal exact-arithmetic lemmas

3.1 Strict floors and equality owners

Definition 3.1. For $x \in \mathbb{R}$, let

$$[x]_{<} := \#\{q \in \mathbb{Z}_{\geq 1} : q < x\}.$$

Thus the bracket counts positive integers under a *strict* cutoff.

Lemma 3.2 (Strict-bracket identity). *For every real x ,*

$$[x]_{<} = \max\{0, [x] - 1\}. \quad (3)$$

In particular, for every integer $m \geq 1$,

$$[m]_{<} = m - 1, \quad [m + \tfrac{1}{2}]_{<} = m.$$

Consequently an integer equality wall is owned by the excluded side.

Proof. If $x \leq 1$, both sides of (3) vanish. If $x > 1$, the positive integers below x are exactly $1, \dots, [x] - 1$. This also proves the two specializations. $\square$

The same lemma handles the two equality conventions used later. If $A + 1/4 = n \in \mathbb{Z}_{\geq 1}$, then $[A + 1/4]_{<} = n - 1$. If $a = m\pi$ in the optical radial decomposition, the strict radial layers are represented by $(N, t) = (m - 1, 1)$, not by $(m, 0)$.

3.2 Odd multiplicities and block caps

Lemma 3.3 (Odd-sum and block-cap identity). *For every integer $h \geq 0$,*

$$\sum_{\ell=0}^h (2\ell + 1) = (h + 1)^2, \quad \sum_{\ell=0}^{h-1} (2\ell + 1) = h^2. \quad (4)$$

More generally, if radial layer n contains a complete angular block $0 \leq \ell \leq h_n$, with $h_n = -1$ denoting an empty block, then its total multiplicity is

$$\sum_n (h_n + 1)^2. \quad (5)$$

If $x^2 = h(h + 1)$ is a strict product equality wall, the equality channel $\ell = h$ is absent and the contribution is h^2 .

Proof. The first identity follows either by induction or by $(\ell + 1)^2 - \ell^2 = 2\ell + 1$. Summing over independent radial layers gives (5). At $x^2 = h(h + 1)$, $\sqrt{x^2 + 1/4} - 1/2 = h$; Lemma 3.2 excludes $\ell = h$, leaving $0 \leq \ell < h$ and hence h^2 . $\square$

For completeness, the cap arithmetic collapsed by this lemma is printed in compact form below. A dash means that another analytic argument has declared the cell impossible; the lemma asserts only the arithmetic of the non-dash entries.

$k_8 : H \setminus h$	$\emptyset$	0	1	2	3
7	64	65	68	73	80
6	49	50	53	—	—
5	36	37	40	45	52
4	25	26	29	34	41

$k_9 : H \setminus h$	$\emptyset$	0	1	2	3	4
8	81	—	—	—	—	—
7	64	65	68	73	—	—
6	49	50	53	58	65	74
5	36	37	40	45	52	61
4	25	26	29	34	41	50

Indeed every entry is $(H+1)^2$ plus 0 or $(h+1)^2$. At k_7 , the first blocks $H = 4, 5, 6$ give 25, 36, 49, and second channels 0, 1 contribute $1 + 3 = 4$. The remaining printed caps decompose as

$$\begin{aligned}
k_{10} : \quad & 10^2; \quad 9^2, 9^2 + 4^2; \\
& 8^2, 8^2 + 4^2, 8^2 + 5^2, 8^2 + 6^2; \\
& 7^2 + 5^2, 7^2 + 4^2 + 2^2, 7^2 + 5^2 + 2^2, 7^2 + 6^2, 7^2 + 6^2 + 2^2 \\
& = 100; \quad 81, 97; \quad 64, 80, 89, 100; \quad 74, 69, 78, 85, 89, \\
k_{11} : \quad & 11^2; \quad 10^2, 10^2 + 5^2; \quad 9^2 + 5^2, 9^2 + 5^2 + 2^2; \quad 8^2 + 5^2 + 2^2 \\
& = 121; \quad 100, 125; \quad 106, 110; \quad 93.
\end{aligned}$$

The conservative cap-74 row is the same identity $7^2 + 5^2 = 74$.

The six lower inventories similarly give, in their printed order,

$$(6^2 + 3^2, 6^2 + 3^2 + 1, 7^2 + 3^2 + 1, 7^2 + 4^2 + 1, 7^2 + 4^2 + 1 + 3, 7^2 + 5^2 + 1 + 3) = (45, 46, 59, 66, 69, 78).$$

Finally, the ten lower- d caps are the successive odd sums

channels	cap	channels	cap
1	1	1 + 3	4
1 + 3 + 5	9	1 + 3 + 5 + 1	10
1 + 3 + 5 + 7	16	1 + 3 + 5 + 7 + 1	17
1 + 3 + 5 + 7 + 9 + 1	26	1 + 3 + 5 + 7 + 9 + 1 + 3	29
1 + 3 + 5 + 7 + 9 + 11 + 1 + 3	40	1 + 3 + 5 + 7 + 9 + 11 + 1 + 3 + 5	45.

3.3 The symbolic optical product deficit

Proposition 3.4 (Product-deficit identity and uniform reserve). *Let $N \geq 1$ be an integer, $0 < t \leq 1$, and $x = N + t$. Then*

$$\frac{2}{3}x^3 - Nx^2 + \sum_{n=1}^N n^2 = \frac{N^2}{2} + N \left(t^2 + \frac{1}{6} \right) + \frac{2t^3}{3}. \quad (6)$$

For $C_D = 1382/3125$, the right-hand side is strictly larger than $C_D(N + t)^2$. More precisely, its difference from $C_D(N + t)^2$ is at least

$$\frac{1822532}{91552734375} > 0. \quad (7)$$

Proof. Insert $\sum_{n=1}^N n^2 = N(N+1)(2N+1)/6$ and expand; this proves (6) for all N, t , rather than at sampled values. Put $C = C_D$, $A = 1/2 - C = 361/6250$, and let $G_N(t)$ denote the difference under consideration. Then

$$G_N(t) = AN^2 + N \left(t^2 - 2Ct + \frac{1}{6} \right) + \frac{2t^3}{3} - Ct^2.$$

Consequently

$$G_{N+1}(t) - G_N(t) = A(2N+1) + t^2 - 2Ct + \frac{1}{6}.$$

Since $0 < C < 1$, the last quadratic has minimum $1/6 - C^2$ on $(0, 1]$. For $N \geq 1$,

$$G_{N+1}(t) - G_N(t) \geq 3A + \frac{1}{6} - C^2 = \frac{4229603}{29296875} > 0.$$

It remains to minimize G_1 . Direct differentiation gives

$$G_1'(t) = 2(t - C)(t + 1),$$

so its unique minimum on $0 < t \leq 1$ is at $t = C$. Exact substitution gives

$$G_1(C) = \frac{1822532}{91552734375} > 0,$$

which proves both assertions. $\square$

3.4 Monotonic propagated cuts

Lemma 3.5 (Zero-cut/channel-cut monotonicity). *Fix $n \geq 2$, a checkpoint m , and $M = m(m+1)$. Let $\mathcal{A}_n$ be a finite collection of anchor pairs $(j, q_{j,n}^2)$, where $j_{j+1/2,n}^2 > q_{j,n}^2$ at each anchor. For every ℓ above at least one anchor, set*

$$\mathfrak{B}_n(\ell) := \max_{(j, q_{j,n}^2) \in \mathcal{A}_n, j \leq \ell} (q_{j,n}^2 + \ell(\ell+1) - j(j+1)), \quad (8)$$

$$\mathfrak{Z}_{m,n}(\ell) := \mathfrak{B}_n(\ell) - M, \quad (9)$$

$$\mathfrak{C}_{m,n}(\ell) := \frac{M - \ell(\ell+1)}{n^2 - 1}. \quad (10)$$

Then

$$\mathfrak{Z}_{m,n}(\ell+1) \geq \mathfrak{Z}_{m,n}(\ell) + 2(\ell+1) > \mathfrak{Z}_{m,n}(\ell), \quad (11)$$

$$\mathfrak{C}_{m,n}(\ell+1) = \mathfrak{C}_{m,n}(\ell) - \frac{2(\ell+1)}{n^2 - 1} < \mathfrak{C}_{m,n}(\ell). \quad (12)$$

In particular, if $\mathfrak{Z}_{m,n}(\ell_0) \geq \mathfrak{C}_{m,n}(\ell_0)$, then no channel (ℓ, n) with $\ell \geq \ell_0$ can satisfy the strict proxy (1) at $K = k_m$.

Proof. Every old candidate in (8) gains exactly $(\ell+1)(\ell+2) - \ell(\ell+1) = 2(\ell+1)$ when ℓ is replaced by $\ell+1$; a new anchor can only increase the maximum. This proves (11). Formula (12) is direct.

If a channel were counted at k_m , its product part would give

$$(n^2 - 1)z^2 < M - \ell(\ell+1), \quad \text{hence} \quad z^2 < \mathfrak{C}_{m,n}(\ell),$$

whereas its propagated ball-zero part would give

$$\mathfrak{B}_n(\ell) < z^2 + M, \quad \text{hence} \quad z^2 > \mathfrak{Z}_{m,n}(\ell).$$

These strict inequalities are incompatible when $\mathfrak{Z} \geq \mathfrak{C}$. Equations (11) and (12) propagate the incompatibility to every larger angular index. $\square$

4 Packet B: moving-3z ordering

We first record rational radical enclosures. All six follow by positive squaring; the exact square margins are

lower comparison	margin	upper comparison	margin
$(183/10)^2 < 337$	211/100	$337 < (92/5)^2$	39/25
$(1051/100)^2 < 7073/64$	2221/40000	$7073/64 < (1052/100)^2$	6191/40000
$(1067/100)^2 < 114$	1511/10000	$114 < (107/10)^2$	49/100.

Thus

$$\frac{183}{10} < \sqrt{337} < \frac{92}{5}, \quad \frac{1051}{100} < \frac{\sqrt{7073}}{8} < \frac{1052}{100}, \quad \frac{1067}{100} < \sqrt{114} < \frac{107}{10}. \quad (13)$$

Lemma 4.1 (Endpoint and fixed-wall order). *With $d = \sqrt{337}/2$,*

$$d < 3z_{\rho_0} < 10 < \frac{81}{8} < \frac{21}{2} < \frac{\sqrt{7073}}{8} < \sqrt{114} < 3z_{\rho_c}. \quad (14)$$

Moreover $(81/8)^2 + 8 = 7073/64$.

Proof. The nontrivial endpoint estimates have the following rational margins. From (2) and (13),

$$3z_{\rho_0} > 3\pi > \frac{999}{106}, \quad d < \frac{46}{5}, \quad \frac{999}{106} - \frac{46}{5} = \frac{119}{530} > 0.$$

Also $\rho_0 < 10/183$, whence

$$3z_{\rho_0} < \frac{66}{7} \frac{183}{173} = \frac{12078}{1211}, \quad 10 - \frac{12078}{1211} = \frac{32}{1211} > 0.$$

At the other endpoint, $z_{\rho_c} = \pi + 1/2$; therefore

$$3z_{\rho_c} > \frac{579}{53}, \quad \sqrt{114} < \frac{107}{10}, \quad \frac{579}{53} - \frac{107}{10} = \frac{119}{530} > 0.$$

The rational middle comparisons are immediate. For the two radical steps one may square positive quantities:

$$\frac{7073}{64} - \left(\frac{21}{2}\right)^2 = \frac{17}{64} > 0, \quad 114 - \frac{7073}{64} = \frac{223}{64} > 0.$$

Finally, $(81/8)^2 + 8 = (6561 + 512)/64 = 7073/64$. $\square$

For a positive wall w , the unique solution of $3z_\rho = w$ is

$$r_w := 1 - \frac{3\pi}{w}.$$

The following table encloses each crossing between explicit rational numbers. The bounds use only Lemma 2.1 and (13); for instance r_w increases with w and decreases with π .

wall w	exact rational enclosure for w	exact enclosure for r_w
10	10	$\frac{2}{35} < r_{10} < \frac{61}{1060}$
81/8	81/8	$\frac{13}{189} < r_{81/8} < \frac{11}{159}$
21/2	21/2	$\frac{5}{49} < r_{21/2} < \frac{38}{371}$
$\sqrt{7073}/8$	$1051/100 < w < 1052/100$	$\frac{73}{757} < r_w < \frac{2903}{27878}$
$\sqrt{114}$	$1067/100 < w < 107/10$	$\frac{79}{679} < r_w < \frac{676}{5671}$

The four gaps between consecutive rational enclosures are, in order,

$$\frac{2251}{200340}, \quad \frac{256}{7791}, \quad \frac{183}{389921}, \quad \frac{231225}{18929162}, \quad (15)$$

and are all positive. For example, the first is $13/189 - 61/1060$, and similarly for the other three. The endpoint margins are also explicit:

$$\rho_0 < \frac{10}{183} < \frac{2}{35}, \quad \frac{2}{35} - \frac{10}{183} = \frac{16}{6405} > 0,$$

and, since $\rho_c > 7/51$,

$$\frac{676}{5671} < \frac{7}{51} < \rho_c, \quad \frac{7}{51} - \frac{676}{5671} = \frac{5221}{289221} > 0.$$

We have therefore proved the complete strict order

$$\rho_0 < r_{10} < r_{81/8} < r_{21/2} < r_{\sqrt{7073}/8} < r_{\sqrt{114}} < \rho_c. \quad (16)$$

This single table replaces the ten crossing-location predicates and the four crossing-order predicates of the executable registry.

5 Packet B: checkpoint boundary table and exhaustiveness

The ball-zero input consists of the following strict rational lower bounds:

$$\frac{n=2, \ell}{q_{\ell,2}^2} \left| \begin{array}{cc} 1 & 2 \\ (77/10)^2 & 9^2 \end{array} \right. \frac{3}{(103/10)^2} \frac{5}{(129/10)^2} \quad \frac{n=3, \ell}{q_{\ell,3}^2} \left| \begin{array}{cc} 1 & 2 \\ (21/2)^2 & (61/5)^2 \end{array} \right. \quad (17)$$

Angular propagation gives the lower bound (8) at all larger angular indices. The seven first-excluded channels have the exact cuts shown below. Here $M = m(m+1)$, $B = \mathfrak{B}_n(\ell_0)$, $Z = B - M$, $C = (M - \ell_0(\ell_0 + 1))/(n^2 - 1)$, and the last column is the decisive incompatibility margin $Z - C$.

m	n	first excluded ℓ_0	B	Z	C	$Z - C$
7	2	2	81	25	50/3	25/3
8	2	4	11409/100	4209/100	52/3	7427/300
9	2	5	16641/100	7641/100	20	5641/100
10	2	6	17841/100	6841/100	68/3	13723/300
11	2	5	16641/100	3441/100	34	41/100
10	3	2	3721/25	971/25	13	646/25
11	3	2	3721/25	421/25	63/4	109/100

Every table entry is reconstructible from (17). The only propagated base values not already anchors are

$$\frac{11409}{100} = \left(\frac{103}{10}\right)^2 + (4 \cdot 5 - 3 \cdot 4), \quad \frac{17841}{100} = \left(\frac{129}{10}\right)^2 + (6 \cdot 7 - 5 \cdot 6).$$

All seven margins are positive; the smallest is $41/100$, so none of these exclusions relies on an equality convention. By Lemma 3.5, each row excludes not merely the displayed channel but its entire angular tail.

It remains to rule out the radial tails. On the high strip $\rho_c \leq \rho < 1$,

$$z \geq z_{\rho_c} = \pi + \frac{1}{2} > \frac{333}{106} + \frac{1}{2} = \frac{193}{53}, \quad z^2 > \frac{37249}{2809}.$$

The exact margins needed below are

$$\frac{37249}{2809} - \frac{90}{8} = \frac{22591}{11236} > 0, \quad \frac{37249}{2809} - \frac{132}{15} = \frac{62649}{14045} > 0. \quad (18)$$

At $m \leq 9$, an $n = 3$ channel would require $8z^2 < M - \ell(\ell + 1) \leq 90$, contradicting the first inequality. At $m \leq 11$, every $n \geq 4$ channel would require $15z^2 \leq (n^2 - 1)z^2 < M - \ell(\ell + 1) \leq 132$, contradicting the second inequality. These are the two base radial exclusions.

Theorem 5.1 (Exhaustive checkpoint inventories). *For $\rho_c \leq \rho < 1$, the channels that can satisfy the strict proxy (1) at the five checkpoints are contained in the following lists:*

checkpoint	possible modes
k_7	first 0:6, second 0:1, no $n \geq 3$
k_8	first 0:7, second 0:3, no $n \geq 3$
k_9	first 0:8, second 0:4, no $n \geq 3$
k_{10}	first 0:9, second 0:5, third 0:1, no $n \geq 4$
k_{11}	first 0:10, second 0:4, third 0:1, no $n \geq 4$.

Here n -th means radial index n , and $a:b$ denotes the angular indices $a, a + 1, \dots, b$. The word “possible” asserts an exhaustive upper inventory, not that every listed channel is present for every ratio.

Proof. For $n = 1$, the product part of the strict proxy at k_m is

$$z^2 + \ell(\ell + 1) < z^2 + m(m + 1).$$

Thus $\ell < m$. The equality channel $\ell = m$ and all larger indices are excluded, with the equality ownership fixed by Lemma 3.2. This gives precisely first 0: $m - 1$.

For $n = 2$, apply the first five rows of the checkpoint table and then Lemma 3.5. They give first-excluded angular indices 2, 4, 5, 6, 5 at $m = 7, 8, 9, 10, 11$, respectively. Hence the second blocks end at 1, 3, 4, 5, 4.

For $n = 3$, the first inequality in (18) removes all channels through k_9 . At k_{10} and k_{11} , the last two rows of the checkpoint table and cut monotonicity remove every $\ell \geq 2$, leaving only $\ell = 0, 1$. Finally, the second inequality in (18) removes all $n \geq 4$ through k_{11} . The five lists follow, without any bounded angular search. $\square$

6 Executable-row coverage contract

The former executable baseline contained 611 rows and had canonical digest

afd7e054f52aa9a3992fc8ef16d3e7551586c76bfb589053714f5f55a36792f1.

The digest is included only to identify the audited baseline. It is not a premise of any proof above. The exhibits above replace disjoint row groups as follows:

analytic exhibit	rows	exact ledger subfamilies
Lemma 2.1	4	one enclosure-nonemptiness row, two rational π -bound rows, and one reciprocal row
Lemma 3.2	24	sixteen integer/half-integer strict brackets, seven phase-proxy integer walls, and one optical radial equality owner
Lemma 3.3	97	twelve strict angular equality blocks, twelve optical angular blocks, 62 high-checkpoint cap identities, one six-cap lower tuple, and ten lower- d cap identities
Proposition 3.4	23	five general product-deficit predicates and eighteen sampled expansions
Lemma 3.5, the checkpoint boundary table, and Theorem 5.1	117	76 repeated cut-monotonicity rows, seven first-excluded comparisons, two radial exclusions, and 32 metadata/tail instances
Lemma 4.1 and the moving-crossing table	23	five fixed-wall orders, one propagation identity, three endpoint comparisons, ten crossing locations, and four crossing-order rows
total	288	

The following family table gives the same accounting against the executable entry points and makes the partial-family boundaries explicit. It maps the exact rows replaced by this note. “Collapsed” means that multiple sampled or bookkeeping rows are consequences of one displayed identity; it does not mean that an unprinted program enumeration is being used.

executable family	family rows	replaced here	analytic replacement
verify_pi	4	4	Lemma 2.1
verify_thresholds	34	23	five fixed-wall orders, one propagation identity, three endpoint comparisons, and fourteen moving-crossing predicates
verify_strict_face_conventions	35	35	Lemmas 3.2 and 3.3
verify_lower_inventories_and_payments	44	1	single six-cap tuple bookkeeping row
verify_checkpoint_inventories	117	117	boundary table, radial margins, and Theorem 5.1
verify_cap_tables_and_payments	98	62	all block-cap arithmetic: 61 cap-table rows plus the cap-74 block identity
verify_optical_constants	50	36	twelve angular blocks, one radial equality owner, five general product-deficit facts, and eighteen sampled expansions
supplemental k_6 /lower- d ledger	26	10	ten lower- d block-cap identities
total		288	

For the checkpoint family, the 117 rows decompose exactly as follows: 32 finite metadata/tail instances, seven first-excluded zero/channel comparisons, two base radial exclusions, and 76 repeated cut-monotonicity instances. Theorem 5.1 replaces all four groups. For the optical family, Proposition 3.4 replaces the 18 sampled (N, t) expansions by a symbolic identity valid for every integer $N \geq 1$ and every $0 < t \leq 1$.

The remaining 323 rows are outside Packets A and B. In particular, this note makes no claim to replace the high-event cap/payment implications, the Bessel sign registry, seam payments, optical screen constants, or the exact owner subtraction. Those require the other analytic replacement packets. This separation prevents a universal arithmetic lemma from being mistaken for proof of a geometric event assignment.

Connected High-Event Registry

Analytic ledger replacement, Packet C

Abstract

This note replaces the exact-executable arithmetic used in the high spherical-shell staircase by displayed rational witnesses. It contains the complete seventeen-face localization registry, the three algebraic splits, the three dashed (impossible) cap cells, the five global checkpoint payments, the seven conditional payments, and the corrected cap-74 path. All comparisons are strict rational comparisons after the two elementary bounds $333/106 < \pi < 22/7$. In particular, no decimal approximation, interval package, or external ledger is a logical input to this packet.

1 Definitions and strict proxy convention

For $0 < \rho < 1$, put

$$z = z_\rho := \frac{\pi}{1-\rho}, \quad x = x_\rho := z_\rho^2, \quad k_m(\rho) := \sqrt{x_\rho + m(m+1)}. \quad (1)$$

The three-dimensional Weyl term is

$$\mathcal{W}(\rho, K) := \frac{2}{9\pi}(1-\rho^3)K^3. \quad (2)$$

At $K = k_m(\rho)$, the product part of the strict channel proxy for the mode (ℓ, n) , $n \geq 2$, is

$$n^2x + \ell(\ell+1) < x + m(m+1).$$

It is therefore governed by

$$\Delta_{m;n,\ell}(\rho) := m(m+1) - \ell(\ell+1) - (n^2-1)x_\rho. \quad (3)$$

The channel is product-possible only when $\Delta_{m;n,\ell} > 0$. Equality is excluded because the spectral proxy is strict. Since

$$x'_\rho = \frac{2\pi^2}{(1-\rho)^3} > 0, \quad \Delta'_{m;n,\ell} = -(n^2-1)x'_\rho < 0, \quad (4)$$

a positive witness at r controls the side $\rho \leq r$, while a negative witness controls the side $\rho \geq r$.

We write H for the largest first-mode angular index retained in a cap cell and h for the largest second-mode angular index. A complete block through H has multiplicity $(H+1)^2$, because $\sum_{\ell=0}^H (2\ell+1) = (H+1)^2$.

2 The only transcendental enclosure

Lemma 1 (Rational enclosure of π). *One has*

$$\pi_- := \frac{333}{106} < \pi < \frac{22}{7} =: \pi_+. \quad (5)$$

Proof. For $0 < u < 1$, the alternating expansion of $\arctan u$ lies between each pair of consecutive partial sums. The tangent addition formula gives

$$\tan\left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239}\right) = 1;$$

indeed, $\tan(4 \arctan(1/5)) = 120/119$, so both the numerator and denominator in the last tangent quotient are $28561/(119 \cdot 239)$. The angle lies in $(0, \pi/2)$, and hence Machin's identity follows:

$$\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}.$$

Using an even partial sum for the positive term and an upper partial sum for the subtracted term gives

$$\begin{aligned} \pi &> 16 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} - \frac{1}{7 \cdot 5^7} \right) - \frac{4}{239}, \\ [\text{right side}] - \frac{333}{106} &= \frac{3418213}{41563593750} > 0. \end{aligned}$$

Likewise,

$$\begin{aligned} \pi &< 16 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} \right) - 4 \left(\frac{1}{239} - \frac{1}{3 \cdot 239^3} \right), \\ \frac{22}{7} - [\text{right side}] &= \frac{1845738322}{1493178640625} > 0. \end{aligned}$$

This proves (5). □

3 The seventeen rational localization faces

Let

$$\rho_c := \frac{1}{1 + 2\pi}.$$

Because $\pi > 3$, $\rho_c < 1/7$. The seventeen registry ratios are strictly ordered as follows:

$$\begin{aligned} \frac{4}{25} &< \frac{1}{5} < \frac{1}{4} < \frac{3}{10} < \frac{1}{3} < \frac{7}{20} < \frac{3}{8} < \frac{2}{5} < \frac{5}{12} < \frac{3}{7} < \frac{1}{2} \\ &< \frac{8}{15} < \frac{107}{200} < \frac{3}{5} < \frac{16}{25} < \frac{13}{20} < \frac{2}{3}. \end{aligned} \tag{6}$$

For a completely explicit distinctness check, the sixteen consecutive gaps in (6) are

$$\frac{1}{25}, \frac{1}{20}, \frac{1}{20}, \frac{1}{30}, \frac{1}{60}, \frac{1}{40}, \frac{1}{40}, \frac{1}{60}, \frac{1}{84}, \frac{1}{14}, \frac{1}{30}, \frac{1}{600}, \frac{13}{200}, \frac{1}{25}, \frac{1}{100}, \frac{1}{60}. \tag{7}$$

Moreover

$$\frac{4}{25} - \frac{1}{7} = \frac{3}{175} > 0, \quad \frac{7}{8} - \frac{2}{3} = \frac{5}{24} > 0.$$

Thus all seventeen ratios lie in the high strip $\rho_c < \rho < 7/8$.

For a positive row below, we use

$$\Delta_{m;n,\ell}(r) > m(m+1) - \ell(\ell+1) - (n^2 - 1) \left(\frac{\pi_+}{1-r} \right)^2.$$

For a negative row, we use

$$-\Delta_{m;n,\ell}(r) > (n^2 - 1) \left(\frac{\pi_-}{1-r} \right)^2 - m(m+1) + \ell(\ell+1).$$

The resulting exact witnesses are the following. “Retain” means only that the product wall does not permit deletion of the indicated block; the independent Bessel wall may still remove it.

no.	r	$(m; n, \ell)$	exact sign witness	one-sided cap/event consequence
1	1/5	(11; 3, 1)	$\Delta > 320/49$	retain the third block 0:1 for $\rho \leq 1/5$
2	3/10	(8; 2, 3)	$-\Delta > 58215/137641$	at k_8 , $h \leq 2$ for $\rho \geq 3/10$
3	1/3	(8; 2, 2)	$-\Delta > 27699/44944$	at k_8 , $h \leq 1$ for $\rho \geq 1/3$
4	3/8	(9; 2, 3)	$\Delta > 2622/1225$	retain the $h = 3$ column below this face
5	2/5	(9; 2, 2)	$\Delta > 248/147$	retain the $h = 2$ column below this face
6	5/12	(9; 2, 1)	$\Delta > 2200/2401$	retain the $h = 1$ column below this face
7	3/7	(9; 2, 0)	$-\Delta > 120843/179776$	no second mode at k_9 on the upper side
8	1/2	(10; 2, 0)	$-\Delta > 23677/2809$	no second mode at k_{10} on the upper side
9	107/200	(11; 2, 0)	$-\Delta > 13302732/2699449$	no second mode at k_{11} on the upper side
10	8/15	(11; 2, 0)	$-\Delta > 2175627/550564$	the clean stated cutoff: a second mode forces $\rho < 8/15$
11	3/5	(8; 2, 0)	$-\Delta > 5080707/44944$	select the no-second k_8 cap/payment side
12	16/25	(9; 2, 0)	$-\Delta > 1555635/11236$	select the no-second k_9 cap/payment side
13	13/20	(10; 2, 0)	$-\Delta > 18126190/137641$	select the no-second k_{10} cap/payment side
14	2/3	(11; 2, 0)	$-\Delta > 1510851/11236$	select the no-second k_{11} cap/payment side
15	4/25	(10; 3, 0)	$-\Delta > 260740/137641$	no third mode at k_{10} on the upper side
16	1/4	(11; 3, 0)	$-\Delta > 23484/2809$	a third mode at k_{11} forces $\rho < 1/4$
17	7/20	(9; 2, 4)	$-\Delta > 36230/474721$	the $H = 6, h = 4$ cap-74 cell can occur only below this face

Every entry is a displayed positive rational number. Together with the monotonicity (4), the table proves the asserted side on an interval, not merely at a sampled point.

4 Three algebraic splits and the five k_{11} implications

At $\rho = \rho_c$, $x = (\pi + 1/2)^2$, while at $\rho = 7/8$, $x = 64\pi^2$. Lemma 1 gives the exact chain

$$(\pi + \tfrac{1}{2})^2 < \left(\frac{51}{14} \right)^2 = \frac{2601}{196} < 16 < \frac{68}{3} < 34 < 64 \cdot 3^2 < 64\pi^2. \quad (8)$$

Consequently each of $x = 16, 68/3, 34$ is attained at a unique point in the high strip. The relevant channel arithmetic is

x	channel	exact value of Δ
16	$(m; n, \ell) = (11; 3, 1)$	$132 - 2 - 8 \cdot 16 = 2 > 0$,
$68/3$	$(10; 2, 6)$	$110 - 42 - 3(68/3) = 0$,
34	$(11; 2, 5)$	$132 - 30 - 3 \cdot 34 = 0$.

(9)

Thus the first split lies on the possible side. The latter two are exact channel walls, and their new modes are excluded by the strict convention.

The two localization consequences at k_{11} may also be read without the table:

$$x_{8/15} - 44 > \left(\frac{\pi_-}{1 - 8/15} \right)^2 - 44 = \frac{725209}{550564} > 0, \quad (10)$$

$$x_{1/4} - \frac{33}{2} > \left(\frac{\pi_-}{1 - 1/4} \right)^2 - \frac{33}{2} = \frac{5871}{5618} > 0. \quad (11)$$

Hence a second mode at k_{11} forces $\rho < 8/15$, and a third mode forces $\rho < 1/4$. Finally, the first-mode ball walls give

$$\left(\frac{69}{5} \right)^2 - 132 = \frac{1461}{25} = 44 + \frac{361}{25} = \frac{33}{2} + \frac{2097}{50}, \quad (12)$$

$$\left(\frac{64}{5} \right)^2 - 132 = \frac{796}{25} = \frac{33}{2} + \frac{767}{50}. \quad (13)$$

If $H = 10$ is present at k_{11} , then x is larger than both the second threshold 44 and the third threshold $33/2$; hence every second and third mode is excluded. If $H = 9$ is present, every third mode is excluded. Equations (10)–(13) are the five formerly separate k_{11} localization implications.

5 The three impossible cap cells

If the first mode H is counted at k_m , its strict ball wall $\beta_{H,1} < k_m$ forces $x > \beta_{H,1}^2 - m(m+1)$. If the second mode h is counted, its product wall forces $x < (m(m+1) - h(h+1))/3$. The three dashed cells are therefore empty by the following exact, incompatible bounds:

cell	required lower bound	required upper bound	gap
$k_8 : H = 6, h \geq 2$	$100 - 72 = 28$	$(72 - 6)/3 = 22$	6
$k_9 : H = 8, h \geq 0$	$(71/6)^2 - 90 = 1801/36$	$90/3 = 30$	$721/36$
$k_9 : H = 7, h \geq 3$	$(23/2)^2 - 90 = 169/4$	$(90 - 12)/3 = 26$	$65/4$.

(14)

No limiting equality can rescue a cell: both necessary modal inequalities are strict.

6 Checkpoint monotonicity and five global payments

Put $M = m(m+1)$ and $\mathcal{W}_m(\rho) := \mathcal{W}(\rho, k_m(\rho))$. Logarithmic differentiation shows that $\mathcal{W}'_m$ has the sign of

$$\pi^2(1 + \rho) - M\rho^2(1 - \rho)^2. \quad (15)$$

For $m \leq 11$,

$$\pi^2(1 + \rho) - M\rho^2(1 - \rho)^2 > 9 - \frac{132}{16} = \frac{3}{4} > 0.$$

Thus every $\mathcal{W}_m$, $m \leq 11$, is strictly increasing. This single argument replaces both executable derivative-bound predicates.

At ρ_c , one has $z = \pi + 1/2 > 193/53$, $\rho_c < 1/7$, and $1/\pi > 7/22$. Consequently

$$\mathcal{W}(\rho_c, K) > \frac{2}{9} \frac{7}{22} \left(1 - \frac{1}{7^3}\right) K^3 = \frac{38}{539} K^3. \quad (16)$$

For each row in the next table, the middle column proves $k_m(\rho_c) > t_m$, and the last column proves $\mathcal{W}_m(\rho_c) > C_m$.

m	cap C_m	t_m	$(193/53)^2 + M - t_m^2$	$(38/539)t_m^3 - C_m$
7	40	208/25	67049/1755625	5083656/8421875
8	53	923/100	1901439/28090000	656778873/269500000
9	73	254/25	61431/1755625	7911557/8421875
10	89	111/10	14211/280900	1999489/269500
11	121	241/20	65271/1123600	5076899/2156000

Every last-column entry is positive. Monotonicity therefore extends each global payment from ρ_c to the complete high strip.

7 Seven conditional first-block payments

We now give a general reduction which avoids any numerical enclosure of a radical. Suppose the presence of first mode H below the checkpoint k_m forces $K > t$. Set

$$a^2 := t^2 - m(m+1), \quad \rho_0 := 1 - \frac{\pi}{a}. \quad (17)$$

All seven rows below have $a > \pi$: the smallest listed radicand is

$$\frac{89}{4}, \quad \frac{89}{4} - \left(\frac{22}{7}\right)^2 = \frac{2425}{196} > 0.$$

If the mode is present for some $K \leq k_m(\rho)$, then $k_m(\rho) > t$, hence $x_\rho > a^2$ and $\rho > \rho_0$. By the strict increase of $\mathcal{W}_m$,

$$\mathcal{W}_m(\rho) > \mathcal{W}_m(\rho_0) = \mathcal{W}(\rho_0, t).$$

Choose a rational $r > \rho_0$. Since $\mathcal{W}(\rho, t)$ decreases with ρ at fixed t , Lemma 1 yields

$$\mathcal{W}(\rho_0, t) > \mathcal{W}(r, t) > \frac{7}{99}(1 - r^3)t^3. \quad (18)$$

The condition $r > \rho_0$ follows, without taking a square root, from

$$L := \pi_-^2 - (1 - r)^2 a^2 > 0. \quad (19)$$

Here is the complete calculation. The column P is the reserve in (18) above the stated cap.

m	H	t	cap	a^2	r	L	$P = \frac{7}{99}(1 - r^3)t^3 - \text{cap}$
7	6	10	53	44	8/15	725209/2528100	18659/2673
8	7	23/2	80	241/4	3/5	64349/280900	213281/49500

m	H	t	cap	a^2	r	L	P
9	8	71/6	81	1801/36	3/5	4713989/2528100	14506973/1336500
10	7	23/2	100	89/4	7/20	2105431/4494400	18539033/6336000
10	8	71/6	97	1081/36	3/7	317585/4955076	2865431/261954
10	9	64/5	100	1346/25	3/5	8811001/7022500	25143284/1546875
11	9	64/5	125	796/25	1/2	536261/280900	58757/12375

All L 's and P 's are positive. Each row therefore proves both the localization of its conditional branch and its Weyl payment.

8 The connected cap-74 path

The seventeenth localization row is tied to the exceptional $k_9, H = 6, h = 4$ cap as follows. Angular propagation from the strict $(3, 2)$ Bessel wall gives

$$j_{9/2,2}^2 > \left(\frac{103}{10}\right)^2 + 20 - 12 = \frac{11409}{100} = 108 + \frac{609}{100}. \quad (20)$$

Accordingly the strict proxy uses $\beta_{4,2}^2 = 11409/100$; the strict inequality belongs to the actual Bessel zero $j_{9/2,2}$, not to the proxy constant. At k_9 , the product wall for $(4, 2)$ is

$$3x < 90 - 20 = 70, \quad \text{i.e.} \quad x < \frac{70}{3}.$$

The ball wall (20), on the other hand, forces

$$x > \frac{11409}{100} - 90 = \frac{2409}{100}, \quad \frac{2409}{100} - \frac{70}{3} = \frac{227}{300} > 0.$$

Thus the fully sharpened cap-74 cell is empty.

For completeness, even the deliberately conservative cell is paid. At $\rho = 7/20$,

$$x_{7/20} - \frac{70}{3} > \frac{400}{169} \left(\frac{333}{106}\right)^2 - \frac{70}{3} = \frac{36230}{1424163} > 0, \quad (21)$$

so a counted $(4, 2)$ mode forces $\rho < 7/20$. Also

$$108 - \left(\frac{207}{20}\right)^2 = \frac{351}{400} > 0, \quad (22)$$

so its ball wall forces $K > 207/20$. Therefore

$$\begin{aligned} \mathcal{W}(\rho, K) &> \frac{7}{99} \left(1 - \left(\frac{7}{20}\right)^3\right) \left(\frac{207}{20}\right)^3 \\ &= \frac{52823261673}{704000000} = 74 + \frac{727261673}{704000000} > 74. \end{aligned} \quad (23)$$

The cap itself is exactly $(6 + 1)^2 + (4 + 1)^2 = 49 + 25 = 74$. Equations (20)–(23) cover both the actual incompatibility and every row of the conservative cap-74 fallback.

9 Exact row map and scope audit

The following map uses the ordering of the two functions in the original exact registry. To keep the table readable, L denotes `verify_localizations`, and C denotes `verify_cap_tables_and_payments`. The map lets a referee match every replaced predicate to a displayed hand calculation.

registry family	row numbers	analytic replacement here
L	1–2	distinctness and connectedness: (6)–(7) and the seventeen-row table
L	3–19	all ratios in the high strip, by $\rho_c < 1/7 < 4/25$ and $2/3 < 7/8$
L	20–36	the seventeen exact sign witnesses in the long table
L	37–42	split location (8) and side/equality arithmetic (9)
L	43–47	the five k_{11} implications (10)–(13)
C	44–55	the twelve non-bookkeeping cap–74 predicates, (20)–(23); row $C56$, the block identity $49 + 25 = 74$, is allocated to Packet A
C	57–59	the three impossible cells, (14)
C	78–79	the general checkpoint monotonicity proof (15)
C	80–84	the five global-payment rows following (16)
C	85–98	the seven localization and seven conditional-payment rows in the final long table

The count reconciliation is

subfamily	connected mathematical objects	registry predicates
registry metadata	1	2
high-strip membership	17	17
localization signs	17	17
algebraic splits	3	6
special k_{11} implications	5	5
cap–74 path	1	12
impossible cells	3	3
checkpoint monotonicity	1	2
global payments	5	5
conditional payments	7	14
total	60	83.

In particular, the five headline Packet–C subfamilies contain exactly

$$17 + 3 + 3 + 5 + 7 = 35$$

connected objects (localization signs, algebraic splits, impossible cells, global payments, and conditional payments). Because each algebraic split has a location and a side predicate, and each conditional payment has a localization and a reserve predicate, these headline objects replace $17 + 6 + 3 + 5 + 14 = 45$ registry predicates. The remaining 38 predicates in the total are the explicitly listed auxiliary closure rows.

Thus this packet replaces all 47 predicates of L and 36 predicates of C . The elementary cap addition rows 1–43, row 56, and rows 60–77 of the latter family are outside the declared Packet C scope; they are instances of the general odd-sum/block-cap identity and are not claimed here. No row in the declared scope is unreconstructed.

Purely Analytic Ledger Replacement, Packet D: Seam Arithmetic and the Two-Sided Payments

Exact hand-checkable exhibit

Abstract

This packet replaces the twenty-two executable seam predicates and the eleven moving/fixed payment predicates used in the high side through k_6 . Every premise is an analytic inequality already derived in the seam or staircase argument, and every remaining decision is displayed as an exact rational identity or strict rational comparison. In particular, the derivative of the seam comparison function is derived explicitly and is bounded by an exact proxy with a positive margin to $16/5$. The six moving k_6 payments are printed with their full rational differences, not merely their numerators. No program execution, floating-point value, or interval calculation is a premise.

1 Rational constants used by both tables

For completeness, the needed bounds for π can be checked from the alternating arctangent series, rather than imported from an executable Machin enclosure. Machin's identity and the alternating remainder rule give

$$\begin{aligned}\pi &> 16 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} - \frac{1}{7 \cdot 5^7} \right) - \frac{4}{239} =: L_\pi, \\ \pi &< 16 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} \right) - 4 \left(\frac{1}{239} - \frac{1}{3 \cdot 239^3} \right) =: U_\pi.\end{aligned}$$

Indeed, the tangent-addition formula proves $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and the displayed truncations have the indicated directions. Direct rational subtraction gives

$$L_\pi - \frac{333}{106} = \frac{3418213}{41563593750} > 0, \quad \frac{1571}{500} - U_\pi = \frac{323354809}{853244937500} > 0. \quad (1)$$

Since

$$\frac{22}{7} - \frac{1571}{500} = \frac{3}{3500} > 0, \quad (2)$$

we may use throughout

$$\underline{\pi} := \frac{333}{106} < \pi < \bar{\pi} := \frac{1571}{500} < \frac{22}{7}. \quad (3)$$

All later occurrences of a rational π bound refer to (3).

2 The twenty-two seam rows

2.1 Analytic input and comparison function

We recall precisely the analytic data to which the finite arithmetic is applied. Put

$$\epsilon = 1 - \rho = y^2, \quad \kappa = K\epsilon^2, \quad 0 < y \leq \frac{1}{\sqrt{6}}, \quad \rho \geq \frac{5}{6}, \quad \kappa \geq 54.$$

For a first plateau of length p , and the quantities m, R in the seam decomposition, set

$$P = py, \quad r = Ry, \quad S = (p + m)y.$$

The angular estimate $d > 5/8$ and $R = p - dm$ imply, whenever $r > 0$,

$$0 < r \leq P, \quad S \leq S_*(P, r) := \frac{13P - 8r}{5}. \quad (4)$$

Equality in the second relation can occur when $m = 0$; no later step requires strictness at this comparison. The curvature-shelf calculation already proved in the seam argument gives the self-consistent inequality

$$P^2 < \frac{2\pi^2 Q}{(1 - \hat{q})^2}, \quad Q = 1 + \frac{y^2}{\kappa} + \frac{yS}{\kappa}, \quad \hat{q} = \frac{y^2}{\rho}(Q - 1). \quad (5)$$

Since the right-hand side increases with Q , define

$$Q_* = 1 + \frac{y^2}{\kappa} + \frac{yS_*}{\kappa}, \quad q_* = a(Q_* - 1), \quad a := \frac{y^2}{\rho} \leq \frac{1}{5}, \quad (6)$$

and

$$H(P, r) := \frac{2\pi^2 Q_*}{(1 - q_*)^2}. \quad (7)$$

Then (4) gives $Q \leq Q_*$ and $\hat{q} \leq q_*$. Since (5) is itself strict and its right-hand side increases with Q , it follows that $P^2 < H(P, r)$.

The remaining arithmetic is exactly the following table. Rows 1–5 are the five constant/shelf rows, rows 6–10 are the five displacement rows, rows 11–14 are the four derivative rows, rows 15–17 are the three endpoint rows, and rows 18–22 are the five terminal-payment rows. Thus the table reconciles one-for-one with all twenty-two seam predicates.

For clarity, the radical used in rows 7–9 follows directly from the analytic shelf bound. Namely,

$$p^2 < \frac{2\pi\rho K}{\epsilon}, \quad P = py, \quad K = \frac{\kappa}{y^4}$$

imply

$$\left(\frac{y^3 P}{\kappa\rho}\right)^2 < \frac{2\pi y^2}{\kappa\rho} \leq \frac{\pi}{135} < \frac{\bar{\pi}}{135}.$$

Together with $S_* \leq 13P/5$, this gives

$$q_* < \frac{1}{1620} + \frac{13}{5} \frac{763}{5000},$$

which is precisely the dependency recorded by rows 6–9.

#	quantity	exact rational check	role
1	seam π bound	$\bar{\pi} - U_\pi = 323354809/853244937500 > 0$	$\pi < \bar{\pi}$; (1)
2	lower $\sqrt{2}$ bound	$2 - (140/99)^2 = 2/9801 > 0$	$140/99 < \sqrt{2}$
3	upper $\sqrt{2}$ bound	$(99/70)^2 - 2 = 1/4900 > 0$	$\sqrt{2} < 99/70$
4	y ceiling	$(49/120)^2 - 1/6 = 1/14400 > 0$	$y < 49/120$
5	shelf ceiling	$(13/5)^2 - 2\pi = 119/250 > 0$	$\sqrt{2\pi\rho} < 13/5$; rows 1

#	quantity	exact rational check	role
6	first displacement term	$\frac{(1/6)^2}{54(5/6)} = \frac{1}{1620}$	$y^4/(\kappa\rho) \leq 1/1620$
7	radical ceiling	$(763/5000)^2 - \frac{\pi}{135} = \frac{8563}{675000000} > 0$	shelf radical; row 1
8	displacement proxy	$\frac{1}{1620} + \frac{13}{5} \frac{763}{5000} = \frac{804689}{2025000}$	rows 6–7
9	displacement margin	$\frac{1620}{400} - \frac{804689}{2025000} = \frac{497}{4050000} > 0$	$q_*, \hat{q} < \bar{q} := 159/400$
10	outer-half localization	$\frac{5}{6}(1 - 159/400) = \frac{241}{480} = \frac{1}{2} + \frac{1}{480}$	$x_0/K > 1/2$; row 9
11	Q_* -derivative proxy	$D_Q := \frac{13}{5} \frac{49/120}{54} = \frac{637}{32400}$	$\partial_P Q_* < D_Q$; row 4
12	exact H -derivative proxy	$\frac{2\pi^2 D_Q (1 + 159/400 + 2/5)}{(1 - 159/400)^3} = \frac{2260740364246}{708624500625}$	rows 1, 9, 11 and $a \leq 1/5$
13	margin to $16/5$	$\frac{16}{5} - \frac{2260740364246}{708624500625} = \frac{6858037754}{708624500625} > 0$	$\partial_P H(P, B) < 16/5$
14	synthetic-slope margin	$2\frac{14}{3} - \frac{16}{5} = \frac{92}{15} > 0$	$\partial_P (P^2 - H) > 92/15$
15	endpoint Q_* proxy	$\bar{Q} := 1 + \frac{1/6}{54} + \frac{(49/120)(14/3)}{54} = \frac{10093}{9720}, \quad \bar{Q} - 1 = \frac{373}{9720}$	$Q_*(B, B) < \bar{Q}$
16	endpoint q_* proxy	$\frac{1}{5}(\bar{Q} - 1) = \frac{373}{48600}$	$q_*(B, B) < 373/48600$
17	initial reserve	$\left(\frac{14}{3}\right)^2 - \frac{2\pi^2(10093/9720)}{(1 - 373/48600)^2} = \frac{2505132463469}{2616573970125} > 0$	$B^2 - H(B, B) > 0$; rows 1, 15–16
18	action-gain coefficient	$g := \frac{2(140/99)54}{3(1571/500)} = \frac{280000}{17281}$	$K\eta > g/y$; rows 1–2
19	gain after R loss	$g - \frac{14}{8} \frac{598066}{99^3} = \frac{51843}{132}$	use $R < 14/(3y)$
20	interface loss proxy	$\frac{8}{15} \frac{99}{70} = \frac{132}{175}$	$4\delta < 132/175$; row 3
21	dangerous-branch reserve	$\frac{598066}{51843} \frac{120}{49} - 1 - \frac{132}{175} = \frac{80132733}{3024175} > 0$	$R > 0$; rows 4, 19–20
22	safe-branch reserve	$\frac{280000}{17281} \frac{120}{49} - 1 - \frac{132}{175} = \frac{114694733}{3024175} > 0$	$R \leq 0$; rows 4, 18, 20

2.2 Why the table proves the seam implications

We spell out the two implications for which the table is more than a collection of arithmetic facts.

First, on the segment $r = B := 14/3$ and $P \geq B$,

$$\partial_P Q_* = \frac{13y}{5\kappa}, \quad \partial_P q_* = a \partial_P Q_*.$$

Differentiation of (7), followed only by the identity $aQ_* = q_* + a$, gives

$$\partial_P H = 2\pi^2(\partial_P Q_*) \frac{1 - q_* + 2aQ_*}{(1 - q_*)^3} = 2\pi^2(\partial_P Q_*) \frac{1 + q_* + 2a}{(1 - q_*)^3}. \quad (8)$$

Rows 1, 9, and 11 and $a \leq 1/5$ substitute monotonically into (8); rows 12–13 then prove $\partial_P H < 16/5$. Because $P \geq B$, row 14 proves

$$\frac{d}{dP}(P^2 - H(P, B)) = 2P - \partial_P H(P, B) > \frac{92}{15} > 0. \quad (9)$$

Rows 15–17 prove $B^2 - H(B, B) > 0$. Integration of (9) therefore yields $P^2 > H(P, B)$ for every $P \geq B$. If $r \geq B$, then $H(P, r) \leq H(P, B)$ because $S_*(P, r)$ decreases with r and H increases with Q_* . This contradicts $P^2 < H(P, r)$. Hence

$$r < B, \quad R < \frac{14}{3y}. \quad (10)$$

Second, the turning-point action bound gives

$$K\eta \geq \frac{2\sqrt{2}\kappa}{3\pi} \frac{1}{y} > \frac{g}{y},$$

where row 18 is the exact rational lower proxy. Also row 4 gives $1/y > 120/49$, and row 20 gives $4\delta < 8\sqrt{2}/15 < 132/175$. With $M = \lfloor K\eta \rfloor - R$ and the strict inequality $\lfloor x \rfloor > x - 1$, (10) and row 21 give, when $R > 0$,

$$M - 4\delta > \left(g - \frac{14}{3}\right) \frac{120}{49} - 1 - \frac{132}{175} > 0.$$

When $R \leq 0$, row 22 gives the still stronger estimate

$$M - 4\delta > g \frac{120}{49} - 1 - \frac{132}{175} > 0.$$

Thus the seam tail has the strict action payment claimed in the main argument, including the $R = 0$ and integer- $K\eta$ faces.

Remark 1 (Notation correction in the executable label). The supplemental verifier labels row 11 as an “exact dq/dP cap.” In the notation of the proof, the number 637/32400 is a proxy for $\partial_P Q_*$, not for $\partial_P q_*$. In fact $\partial_P q_* = a\partial_P Q_*$ and hence $\partial_P q_* < 637/162000$. Formula (8) and the verifier both use 637/32400 in its correct role as the Q_* -derivative proxy. The proof is therefore sound, but the old ledger label is inaccurate.

Remark 2 (Source wording for the π bound). One sentence in the seam passage groups $\pi < 1571/500$ with bounds said to be proved by positive rational squaring. Squaring proves rows 2–5, but not a bound for π . Equations (1)–(3) supply the required analytic alternating-series proof.

3 The eleven k_6 payment rows

Put

$$z_\rho = \frac{\pi}{1 - \rho}, \quad k_m(\rho)^2 = z_\rho^2 + m(m + 1), \quad p_1(\rho)^2 = 4z_\rho^2 + 2, \quad (11)$$

and

$$\mathcal{W}(\rho, K) = \frac{2}{9\pi}(1 - \rho^3)K^3. \quad (12)$$

For a moving wall $T^2 = az_\rho^2 + b$, direct differentiation shows that $\mathcal{W}(\rho, T)$ increases whenever

$$a\pi^2(1 + \rho) > b\rho^2(1 - \rho)^2. \quad (13)$$

Here $b/a \leq 42$. Since $\rho^2(1 - \rho)^2 \leq 1/16$ and $\pi > 3$, the left side of (13), after division by a , is greater than 9, whereas the right side is at most $42/16$; hence every moving payment in the table is increasing. At a fixed wall L , the function $\mathcal{W}(\rho, L)$ is decreasing in ρ .

For a rational split r , define

$$A(a, b; r) := a \frac{(333/106)^2}{(1 - r)^2} + b, \quad B(r) := \frac{7}{99}(1 - r^3). \quad (14)$$

The bounds (3) give, on a moving wall at $\rho = r$,

$$\mathcal{W}(r, T) > B(r)A(a, b; r)^{3/2}. \quad (15)$$

Thus, for a positive cap C , the rational inequality

$$\Delta_{\text{mov}} := B(r)^2 A(a, b; r)^3 - C^2 > 0 \quad (16)$$

implies the strict payment, because

$$\Delta_{\text{mov}} = (B(r)A^{3/2} - C)(B(r)A^{3/2} + C)$$

and the second factor is positive. At a fixed wall L , it is enough to check the ordinary rational reserve

$$\Delta_{\text{fix}} := B(r)L^3 - C > 0. \quad (17)$$

The following is the complete eleven-row registry: six moving rows and five fixed rows. The fractions in the last column are the full reduced differences in (16) or (17).

wall/branch	split r	wall data	cap C	full exact rational difference
k_1 moving	1/8	$(a, b) = (1, 2)$	4	$\Delta_{\text{mov}} = \frac{153864824754340538785}{348796101192617852928} > 0$
k_2 moving	3/10	$(a, b) = (1, 6)$	9	$\Delta_{\text{mov}} = \frac{1399810848073457853053}{394237491401523000000} > 0$
k_3 moving	1/2	$(a, b) = (1, 12)$	16	$\Delta_{\text{mov}} = \frac{137027505353736631}{514922437748928} > 0$
k_4 moving	1/2	$(a, b) = (1, 20)$	25	$\Delta_{\text{mov}} = \frac{2507119282010251109}{13902905819221056} > 0$
k_5 moving	13/25	$(a, b) = (1, 30)$	36	$\Delta_{\text{mov}} = \frac{92672404686738742434741}{709259556128000000000} > 0$
p_1 moving	1/5	$(a, b) = (4, 2)$	29	$\Delta_{\text{mov}} = \frac{373255772037478993002833}{868931613701316000000} > 0$
k_2 fixed	3/10	$L = 51/10$	9	$\Delta_{\text{fix}} = \frac{1387329}{11000000} > 0$
k_3 fixed	1/2	$L = 13/2$	16	$\Delta_{\text{fix}} = \frac{6277}{6336} > 0$
k_4 fixed	1/2	$L = 15/2$	25	$\Delta_{\text{fix}} = \frac{775}{6336} > 0$
k_5 fixed	13/25	$L = 17/2$	36	$\Delta_{\text{fix}} = \frac{704}{452843} > 0$
p_1 fixed	1/5	$L = 77/10$	29	$\Delta_{\text{fix}} = \frac{343750}{849901} > 0$

We now verify that the table pays every branch rather than merely listing positive numbers. For each maximum entry wall

$$\max\{k_2, 51/10\}, \quad \max\{k_3, 13/2\}, \quad \max\{k_4, 15/2\}, \quad \max\{k_5, 17/2\}, \quad \max\{p_1, 77/10\},$$

partition the ratio interval at the displayed r . On $\rho \geq r$, the maximum is at least the moving component and monotonicity together with the moving row pays C . On $\rho \leq r$, the maximum is at least the fixed component and the fixed row pays C . This argument does not assume that the two components are equal at the rational split. For k_1 , the applicable ratio interval starts at $\rho_c > 1/8$ (because $\pi < 22/7 < 7/2$), so the first moving row applies directly. Strict payments also cover coincident entry walls, while the wall itself remains assigned to the lower cap by the strict spectral convention.

4 Reconciliation with the lower- d rows

The two rational π predicates used by the two-sided ledger are already proved in (3). The eleven k_6 predicates are exactly the eleven rows in the preceding table. The remaining thirteen wall/cap predicates in that ledger are the ten transparent multiplicity sums

$$\begin{aligned}
& (1, 1+3, 1+3+5, 1+3+5+1, 1+3+5+7, \\
& \quad 1+3+5+7+1, 1+3+5+7+9+1, \\
& \quad 1+3+5+7+9+1+3, \\
& \quad 1+3+5+7+9+11+1+3, \\
& \quad 1+3+5+7+9+11+1+3+5) \\
& \quad = (1, 4, 9, 10, 16, 17, 26, 29, 40, 45),
\end{aligned} \tag{18}$$

and the endpoint ordering $13/2 < 2z_\rho < 15/2$. The latter follows without an executable check: using $\sqrt{337} < 19$ on the lower endpoint and (3),

$$2 \frac{333}{106} \frac{19}{18} - \frac{13}{2} = \frac{7}{53} > 0, \quad \frac{15}{2} - \left(2 \frac{22}{7} + 1\right) = \frac{3}{14} > 0. \tag{19}$$

Thus the cap-10 row is empty, and the three executable ordering rows collapse to (19) and its immediate empty-cell consequence.

The nine lower- d payment rows need no new table here: the main paper’s lemma “Lower side through d ” (label `cm:lem-lower-d`) already prints the seven fixed-wall differences

$$\frac{793292}{1546875}, \frac{486236661}{1375000000}, \frac{333100003}{99000000}, \frac{329797}{88000}, \frac{3590476297}{1125000000}, \frac{327078887}{99000000}, \frac{8805519}{1375000},$$

and states the initial and moving payments $\mathcal{W}(\rho_c, L(\rho_c)) > 19/6$, where $L(\rho) = (2\rho)^{-1}$ on this branch, and $\mathcal{W}(1/19, 2z) > 508/27$. If desired, their exact printed lower proxies close by the two one-line subtractions

$$\frac{57421}{16854} - \frac{19}{6} = \frac{675}{2809} > 0, \quad \frac{173863}{8427} - \frac{508}{27} = \frac{137795}{75843} > 0. \tag{20}$$

Consequently every seam and two-sided payment predicate assigned to Packet D is now visible as analytic reasoning plus printed integer arithmetic.

5 Dependency count and conclusion

The replacement count is exact:

$$\underbrace{22}_{\text{seam table}} + \underbrace{(2+6+5+10+3)}_{k_6/\text{lower-}d \text{ wall-cap ledger}} + \underbrace{9}_{\text{already printed lower-}d \text{ payments}}.$$

The two constant rows are supplied by (3); the six plus five payment rows are the eleven-row table; the ten bookkeeping rows are (18); the three ordering rows are (19); and the nine already printed payments are explicitly cross-referenced above. This removes program execution from the logical dependency of precisely the Packet-D portion of the exact ledger.

Upper-Owner and Optical Cleanup

Analytic ledger replacement, Packet E

Abstract

This note replaces the executable scope checks surrounding the upper owner of the final spherical-shell residual. It proves directly that the eligible upper wall is $K_0(\rho)$ and that $k_{11}(\rho) < K_0(\rho)$ on the required ratio interval. It also records the symbolic identities that make the repeated optical ledger samples unnecessary.

1 The upper owner

For $0 < \rho < 1$, set

$$a_\rho := \frac{2\pi\rho}{1-\rho}, \quad \eta_\rho := G_1(\max\{\rho, 1/2\}),$$

where

$$G_1(s) = \frac{\sqrt{1-s^2} - s \arccos s}{\pi}, \quad \omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}.$$

Let $C_0 > 0$, and define

$$K_0(\rho) := \left(\frac{\sqrt{a_\rho} + \sqrt{a_\rho + 4\eta_\rho C_0}}{2\eta_\rho} \right)^2. \quad (1)$$

Also put

$$\rho_c := \frac{1}{1+2\pi}, \quad z_\rho := \frac{\pi}{1-\rho}, \quad k_{11}(\rho) := \sqrt{z_\rho^2 + 132}.$$

Lemma 1 (The final upper wall is genuinely above the staircase). *For*

$$\rho_c \leq \rho < \frac{7}{8},$$

one has

$$k_{11}(\rho) < 64 < U(\rho). \quad (2)$$

More precisely, $U(\rho) = K_0(\rho) > 64$ for $\rho_c \leq \rho < 5/6$, while

$$U(\rho) = \frac{54}{(1-\rho)^2} \geq 1944 \quad (5/6 \leq \rho < 7/8).$$

In particular, on the final surviving interval $\rho_c \leq \rho < 39/50$, the upper owner is $K_0(\rho)$.

Proof. We use only

$$3 < \pi < \frac{22}{7}, \quad \sqrt{3} < \frac{7}{4}.$$

They imply

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6} < \frac{7/4}{6} - \frac{1}{6} = \frac{1}{8}.$$

Since

$$G_1'(s) = -\frac{\arccos s}{\pi} < 0 \quad (0 \leq s < 1), \quad G_1(1) = 0,$$

one has $0 < \eta_\rho \leq \omega_0 < 1/8$. The function a_ρ is increasing, and its value at ρ_c is exactly one:

$$a_{\rho_c} = \frac{2\pi/(1+2\pi)}{2\pi/(1+2\pi)} = 1.$$

Thus $a_\rho \geq 1$. Formula (1) and $C_0 > 0$ give

$$\sqrt{K_0(\rho)} > \frac{\sqrt{a_\rho}}{\eta_\rho} > 8,$$

so $K_0(\rho) > 64$.

On the other hand, $\rho < 7/8$, and therefore

$$z_\rho < 8\pi < \frac{176}{7} < 26.$$

Consequently

$$k_{11}(\rho)^2 < 26^2 + 132 = 808 < 64^2,$$

which proves (2).

The small-hole wall H_0 is eligible only when $\rho < \omega_0$. But

$$\rho_c > \frac{1}{8} > \omega_0,$$

because $\pi < 22/7 < 7/2$ implies $1 + 2\pi < 8$. Also $\pi > 3$ gives the exact scope chain

$$\rho_c < \frac{1}{7} < \frac{39}{50} < \frac{5}{6}.$$

For $\rho < 5/6$, the thin-seam wall is not eligible either, so the piecewise upper-owner formula gives $U = K_0 > 64$. For $5/6 \leq \rho < 7/8$, the seam branch owns the equality face and

$$U(\rho) = \frac{54}{(1-\rho)^2} \geq \frac{54}{(1/6)^2} = 1944 > 64.$$

Combining this with $k_{11} < 64$ proves (2). Finally $39/50 < 5/6$, so the final surviving interval lies wholly in the K_0 branch. $\square$

2 Symbolic optical identities

The optical ledger evaluates the same algebraic statements at several sample values. The following identities replace those repetitions.

For $a/\pi = N + t$, with $N \geq 1$ and $0 < t \leq 1$, the exact continuous-minus-product deficit is

$$\frac{D(a)}{\pi^2} = \frac{N^2}{2} + N \left(t^2 + \frac{1}{6} \right) + \frac{2t^3}{3}. \quad (3)$$

If $C_D = 1382/3125$, direct subtraction gives

$$\frac{D(a)}{\pi^2} - C_D(N+t)^2 = \left(\frac{1}{2} - C_D\right) N^2 + N \left(t^2 - 2C_D t + \frac{1}{6}\right) + \frac{2t^3}{3} - C_D t^2. \quad (4)$$

The coefficient of N^2 is positive. The difference between the right sides at $N+1$ and N is

$$2 \left(\frac{1}{2} - C_D\right) N + \left(\frac{1}{2} - C_D\right) + t^2 - 2C_D t + \frac{1}{6}.$$

The quadratic $t^2 - 2C_D t + 1/6$ has minimum $1/6 - C_D^2$ at $t = C_D$. Consequently, for $N \geq 1$, the displayed difference is at least

$$3 \left(\frac{1}{2} - C_D\right) + \frac{1}{6} - C_D^2 = \frac{4229603}{29296875} > 0.$$

Therefore the deficit increases in N , and it is enough to check $N = 1$. At $N = 1$, differentiation gives

$$\frac{d}{dt} \left(\frac{D(a)}{\pi^2} - C_D(1+t)^2 \right) = 2(t - C_D)(t+1).$$

Thus the unique minimum on $0 < t \leq 1$ occurs at $t = C_D$, where the exact value is

$$\frac{1822532}{91552734375} > 0. \quad (5)$$

This proves the product screen simultaneously for every N, t .

Likewise, the twelve repeated angular block checks are all instances of

$$\sum_{\ell=0}^{m-1} (2\ell+1) = m^2 \quad (m \in \mathbb{N}_0), \quad (6)$$

proved by subtracting consecutive squares. At a radial equality wall the strict convention excludes the newly proposed layer, so no boundary sample is added. These symbolic statements cover every repeated optical row; the two exact endpoint reserves remain those printed in the main optical lemma.

3 The two optical screens with all scaling shown

For completeness, we now print the algebra that was compressed in the main optical lemma. Set

$$\varepsilon_0 = \frac{11}{50}, \quad c = \frac{1126}{625}, \quad q = \frac{106}{333}, \quad C_D = \frac{1382}{3125}.$$

The inequalities

$$\pi > \frac{333}{106}, \quad \pi < \frac{22}{7}$$

give $q > 1/\pi$, and $0 < \varepsilon \leq \varepsilon_0 < 1/2$.

Lemma 2 (Complete optical screen arithmetic). *Let $a = \varepsilon K$.*

1. *If $\pi < a \leq c/\varepsilon$, then*

$$C_D > \frac{2cq}{3} + \frac{\varepsilon}{4} + \varepsilon^2 q^2. \quad (L)$$

2. If $a \geq c/\varepsilon$, define

$$L(\varepsilon) := \frac{(1-\varepsilon)^2}{4} - \frac{(22/7)(1-\varepsilon)\varepsilon}{4c},$$

$$A(\varepsilon) := q\varepsilon + \frac{\varepsilon^2}{4c} + \frac{q\varepsilon^3}{4c} + \frac{\varepsilon^4}{16c^2}.$$

Then $L(\varepsilon) > A(\varepsilon)$.

3. The low and high domains are both closed at $a = c/\varepsilon$, so their union covers every $a > \pi$.

Proof. For the low screen, $D(a) > C_D a^2$ was proved above. The sufficient product inequality obtained in the main optical argument is

$$D(a) \geq \varepsilon \left(\frac{2a^3}{3\pi} + \frac{a^2}{4} \right) + \frac{\varepsilon^2 a}{\pi}. \quad (1)$$

After division by a^2 , the three terms on the right are bounded by

$$\frac{2\varepsilon a}{3\pi} \leq \frac{2cq}{3}, \quad \frac{\varepsilon}{4} \leq \frac{\varepsilon_0}{4}, \quad \frac{\varepsilon^2}{\pi a} < \frac{\varepsilon^2}{\pi^2} < \varepsilon_0^2 q^2.$$

Exact subtraction gives

$$C_D - \left(\frac{2cq}{3} + \frac{11}{200} + \left(\frac{11}{50} \right)^2 q^2 \right) = \frac{39569}{2772225000} > 0. \quad (7)$$

This proves (L). For $0 \leq a \leq \pi$, the strict radial count is zero, including $a = \pi$, so no low-frequency gap remains.

For the high screen, the exact radial deficit and angular ceiling error from the main analytic argument are

$$\frac{D_{\text{rad}}}{a^2} \geq \frac{(1-\varepsilon)^2}{4} - \frac{\pi(1-\varepsilon)}{4a},$$

$$\frac{E}{a^2} = \frac{\varepsilon}{\pi} + \frac{\varepsilon^2}{4\pi a} + \frac{\varepsilon}{4a} + \frac{\varepsilon^2}{16a^2}.$$

Since $a \geq c/\varepsilon$, one has $1/a \leq \varepsilon/c$. Using the two rational bounds for π therefore gives

$$\frac{D_{\text{rad}}}{a^2} \geq L(\varepsilon), \quad \frac{E}{a^2} < A(\varepsilon).$$

Moreover,

$$L'(\varepsilon) = -\frac{1-\varepsilon}{2} - \frac{22/7}{4c}(1-2\varepsilon) < 0 \quad (0 < \varepsilon < 1/2),$$

whereas $A'(\varepsilon) > 0$ coefficientwise. Hence

$$L(\varepsilon) - A(\varepsilon) \geq L(\varepsilon_0) - A(\varepsilon_0).$$

The endpoint subtraction is

$$L(\varepsilon_0) - A(\varepsilon_0) = \frac{1}{4} \left(\frac{39}{50} \right)^2 - \frac{22}{7} \frac{(39/50)(11/50)}{4(1126/625)}$$

$$- \left[\frac{11}{50} q + \frac{(11/50)^2}{4(1126/625)} + \frac{(11/50)^3 q}{4(1126/625)} + \frac{(11/50)^4}{16(1126/625)^2} \right] \quad (8)$$

$$= \frac{14817541}{472867032960000} > 0.$$

Thus the radial deficit strictly exceeds the angular error.

Finally,

$$\frac{c}{\varepsilon} \geq \frac{c}{\varepsilon_0} = \frac{2252}{275} > \frac{22}{7} > \pi, \quad \frac{2252}{275} - \frac{22}{7} = \frac{9714}{1925} > 0.$$

Therefore the common face lies in the low domain as well as the high domain, and

$$\{a : \pi < a \leq c/\varepsilon\} \cup \{a : a \geq c/\varepsilon\} = \{a : a > \pi\},$$

and equality $a = c/\varepsilon$ belongs to both pieces. Together with the zero-count interval $0 \leq a \leq \pi$, this covers every $a \geq 0$. At $K = a = 0$, both the spectral count and the Weyl term vanish. $\square$

Ledger rows replaced. Lemma 1 replaces the eight executable `verify_u_and_k11` scope checks. Equations (3)–(6) replace the repeated product samples and odd-block identities. These are rows of `verify_optical_constants`. Lemma 2 prints the remaining scaling, monotonicity, common-face, and exact-reserve implications. Thus all fifty optical rows are consequences of displayed identities rather than executable predicates.

Exact Owner Subtraction without a State Enumeration

Analytic ledger replacement, Packet F

Abstract

The executable exact ledger checks a 9×7 truth table when the accepted lower, staircase, and optical regions are subtracted from the Round-19 residual. This note replaces all sixty-three state checks and the five additional interface checks by one set-theoretic calculation. Every open and closed face is retained explicitly.

1 Sets and inherited owners

Let

$$0 < \rho_* < \rho_0 < \rho_c < \rho_{\text{opt}} < \frac{7}{8}, \quad \rho_{\text{opt}} := \frac{39}{50},$$

and let $U(\rho)$, $k_6(\rho)$, and $k_{11}(\rho)$ be the upper owner and the two staircase walls from the main proof. Put

$$D_{\text{low}} := \left\{ \rho_* < \rho \leq \rho_0, (2\rho)^{-1} < K < U(\rho) \right\} \cup \left\{ \rho_0 < \rho < \rho_c, d < K < U(\rho) \right\}, \quad (1)$$

$$D_{\text{high}} := \left\{ \rho_c \leq \rho < \frac{7}{8}, k_6(\rho) < K < U(\rho) \right\}, \quad (2)$$

where $d = (2\rho_0)^{-1}$, and define

$$D_{19} := D_{\text{low}} \dot{\cup} D_{\text{high}}. \quad (3)$$

The dot is justified by the disjoint ratio ranges.

The already-proved inclusive owners used in the subtraction are:

$$L_{\text{own}} := D_{\text{low}}, \quad (4)$$

$$S_{\text{own}} := \left\{ \rho_c \leq \rho < \frac{7}{8}, k_6(\rho) < K \leq k_{11}(\rho) \right\}, \quad (5)$$

$$O_{\text{own}} := \left\{ \rho_{\text{opt}} \leq \rho < \frac{7}{8}, K \geq 0 \right\}. \quad (6)$$

The lower owner includes the complete $\rho = \rho_0$ fibre. The staircase includes its upper wall $K = k_{11}(\rho)$. The optical owner includes its lower ratio face $\rho = \rho_{\text{opt}}$. In contrast, D_{19} already excludes $K = U(\rho)$.

2 The subtraction

Proposition 1 (Exact residual identity). *One has*

$$\boxed{D_{19} \setminus (L_{\text{own}} \cup S_{\text{own}} \cup O_{\text{own}}) = D_{20}}, \quad (7)$$

where

$$D_{20} := \{\rho_c \leq \rho < \rho_{\text{opt}}, \ k_{11}(\rho) < K < U(\rho)\}. \quad (8)$$

Proof. Since $L_{\text{own}} = D_{\text{low}}$ and the union in (3) is disjoint,

$$D_{19} \setminus L_{\text{own}} = D_{\text{high}}. \quad (9)$$

Inside D_{high} , the lower frequency inequality is strict: $k_6(\rho) < K$. Removing the inclusive staircase (5) therefore gives exactly

$$D_{\text{high}} \setminus S_{\text{own}} = \left\{ \rho_c \leq \rho < \frac{7}{8}, \ k_{11}(\rho) < K < U(\rho) \right\}. \quad (10)$$

There is no missing equality case: $K = k_{11}(\rho)$ belongs to S_{own} , while $K = U(\rho)$ never belonged to D_{19} .

Finally, subtracting the inclusive ratio owner (6) changes $\rho < 7/8$ in (10) to $\rho < \rho_{\text{opt}}$. Its equality face is removed because $\rho = \rho_{\text{opt}}$ belongs to O_{own} . The face $\rho = \rho_c$ remains, since it belongs neither to D_{low} nor to O_{own} . The result is precisely (8). $\square$

3 The five delicate interfaces

For clarity, the five extra assertions formerly checked separately by the executable ledger are consequences of the displayed sets:

1. At $\rho = \rho_0$, the identity $(2\rho_0)^{-1} = d$ makes the prospective second lower fibre empty; the first set in (1) owns the full equality fibre, and L_{own} removes it exactly once.
2. The face $\rho = \rho_c$ belongs to D_{high} . It is not an optical face, so points with $k_{11}(\rho_c) < K < U(\rho_c)$ remain in D_{20} .
3. The face $\rho = \rho_{\text{opt}} = 39/50$ belongs to O_{own} and is therefore absent from D_{20} .
4. The frequency face $K = k_{11}(\rho)$ belongs to S_{own} , whereas $K = U(\rho)$ is excluded already by the strict inequality in D_{19} .
5. If $\rho_c < \rho < \rho_{\text{opt}}$ and $k_{11}(\rho) < K < U(\rho)$, the point is in D_{high} but in none of $L_{\text{own}}, S_{\text{own}}, O_{\text{own}}$; hence it survives, and every surviving point has exactly these properties.

4 Final compact/tail split

Define

$$C_{21} := D_{20} \cap \{K \leq 200\}, \quad T_{21} := D_{20} \cap \{K > 200\}. \quad (11)$$

The elementary partition $(0, \infty) = \{K \leq 200\} \dot{\cup} \{K > 200\}$ gives

$$D_{20} = C_{21} \dot{\cup} T_{21}, \quad D_{20} \setminus (C_{21} \cup T_{21}) = \emptyset. \quad (12)$$

Thus $K = 200$ is assigned uniquely to the compact part, regardless of whether $U(\rho)$ lies below, at, or above 200.

Ledger rows replaced. Proposition 1 replaces all $9 \cdot 7 = 63$ state evaluations in the executable family `verify_exact_subtraction`. The five items above replace its five separately asserted interface predicates. Equation (12) supplies the subsequent compact/tail subtraction without a classifier or executable truth table.

A Structural Reduction for the Compact Shell Proxy

Round 1 analytic note

Abstract

This note isolates a possible purely analytic replacement for the compact interval certificate in the spherical-shell argument. The weighted proxy is first written as an exact layer-cake sum. Its deficit from the action integral is then decomposed into a radial shifted-quadrature deficit and an angular rounding defect. A Stieltjes calculation retains two nonnegative remainders that are discarded by the existing optical estimate. One of those remainders admits a smooth explicit lower bound. The result is a single concrete inequality, displayed in (37), whose proof would close the actual compact residual without numerical subdivision. That final inequality is not proved here.

1 Action and the compact target

For $a > 0$ and $z \geq 0$, put

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z < a, \\ 0, & z \geq a. \end{cases} \quad (1)$$

Fix $0 < \rho < 1$ and $K > 0$, and define

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z), \quad T := A(0) = \frac{(1-\rho)K}{\pi}. \quad (2)$$

The strict positive-integer count is

$$[x]_{<} := \#\{n \in \mathbb{N} : n < x\}.$$

The compact proxy and its continuous payment are

$$P(\rho, K) := \sum_{\substack{\nu=\ell+1/2 < K \\ \ell \in \mathbb{N}_0}} 2\nu [A(\nu) + \tfrac{1}{4}]_{<}, \quad (3)$$

$$W(\rho, K) := \int_0^K 2zA(z) dz = \frac{2(1-\rho^3)K^3}{9\pi}. \quad (4)$$

The actual compact residual is contained in

$$\mathcal{R}_c := \left\{ (\rho, K) : \frac{1}{1+2\pi} \leq \rho \leq \frac{39}{50}, \quad k_{11}(\rho) \leq K \leq 200 \right\}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}. \quad (5)$$

Thus it is enough to prove $P < W$ on $\mathcal{R}_c$.

For $0 < z < K$, direct differentiation gives

$$-A'(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \frac{z}{K}, & \rho K \leq z < K. \end{cases} \quad (6)$$

In particular, A is strictly decreasing from T to 0. Let

$$R : [0, T] \longrightarrow [0, K]$$

be its decreasing inverse, and set

$$F(t) := R(t)^2, \quad g(t) := -F'(t), \quad \tau := A(\rho K) = KG_1(\rho). \quad (7)$$

The derivative in (6) increases in magnitude on $(0, \rho K)$ and decreases in magnitude on $(\rho K, K)$. The inverse curvature has the related, but slightly stronger, property

$$g \text{ decreases on } (0, \tau), \quad g \text{ increases on } (\tau, T). \quad (8)$$

Indeed, on the outer branch, if $z = R(t)$ then

$$g(t) = \frac{2\pi z}{\arccos(z/K)};$$

the quotient increases with z , while z decreases with t . On the inner branch, put

$$\delta(z) := \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}.$$

Differentiation with respect to the intermediate ratio gives

$$\frac{\delta(z)}{z} = \int_{\rho}^1 \frac{ds}{s\sqrt{s^2 K^2 - z^2}}.$$

This integral increases with z . Thus $z/\delta(z)$ decreases with z , and $g(t) = 2\pi z/\delta(z)$ increases as t increases and z decreases. The endpoint and interface data are

$$F(\tau) = \rho^2 K^2, \quad g(\tau) = \frac{2\pi\rho K}{\arccos \rho}, \quad g(T) = \frac{2\pi\rho K}{1 - \rho}. \quad (9)$$

Here and below $g(T)$ is the one-sided limit. Near $t = 0$, one has $g(t) = O(t^{-1/3})$, so all improper Stieltjes boundary terms used below vanish.

2 Exact layer cake and angular rounding

Let

$$x_n := n - \frac{1}{4}, \quad M(r) := \# \left\{ \ell \in \mathbb{N}_0 : \ell + \frac{1}{2} < r \right\} = \max \left\{ 0, \left\lceil r - \frac{1}{2} \right\rceil \right\}. \quad (10)$$

Lemma 1 (Exact layer cake). *For every $0 < \rho < 1$ and $K > 0$,*

$$P(\rho, K) = \sum_{\substack{n \geq 1 \\ x_n < T}} M(R(x_n))^2, \quad W(\rho, K) = \int_0^T F(t) dt. \quad (11)$$

Both identities retain the strict convention at every radial and angular wall.

Proof. For a fixed half-integer ν ,

$$\left[A(\nu) + \frac{1}{4} \right]_{<} = \#\{n \geq 1 : x_n < A(\nu)\}.$$

Since A and R are decreasing, $x_n < A(\nu)$ is equivalent to $\nu < R(x_n)$. Interchanging the two finite sums and using

$$\sum_{\ell=0}^{m-1} (2\ell + 1) = m^2$$

proves the first identity. At $R(x_n) = m + 1/2$, the strict angular count is m , exactly as specified by M .

For the second identity, the area formula for a decreasing inverse gives

$$\int_0^T R(t)^2 dt = \int_0^K 2zA(z) dz.$$

The last integral is (4). □

For every active n , abbreviate

$$r_n := R(x_n), \quad m_n := M(r_n), \quad \alpha_n := r_n - m_n + \frac{1}{2}. \quad (12)$$

The strict convention implies

$$m_n - \frac{1}{2} < r_n \leq m_n + \frac{1}{2}, \quad 0 < \alpha_n \leq 1. \quad (13)$$

This includes $m_n = 0$. Define the angular rounding defect

$$\mathcal{E}_{\text{ang}}(\rho, K) := \sum_{x_n < T} (m_n^2 - r_n^2). \quad (14)$$

The defect of a single radial layer is exactly

$$m_n^2 - r_n^2 = (1 - 2\alpha_n)r_n + \left(\alpha_n - \frac{1}{2}\right)^2. \quad (15)$$

Unlike the one-sided estimate $m_n^2 - r_n^2 < r_n + 1/4$, formula (15) retains the cancellation when $\alpha_n > 1/2$.

Combining (11) and (14) gives the first exact two-term decomposition:

$$W - P = \mathcal{D}_{\text{rad}} - \mathcal{E}_{\text{ang}}, \quad \mathcal{D}_{\text{rad}} := \int_0^T F(t) dt - \sum_{x_n < T} F(x_n). \quad (16)$$

3 The radial deficit with retained positive remainders

Define the shifted counting function and its two primitives by

$$C(t) := \#\{n \geq 1 : x_n < t\}, \quad w(t) := t - C(t), \quad \mathfrak{W}(t) := \int_0^t w(s) ds, \quad \Psi(t) := \mathfrak{W}(t) - \frac{t}{4}. \quad (17)$$

On $0 \leq t \leq 3/4$,

$$\Psi(t) = \frac{1}{2} \left(t - \frac{1}{4} \right)^2 - \frac{1}{32}. \quad (18)$$

For $t = x_n + u$, $0 \leq u \leq 1$,

$$\Psi(t) = \frac{1}{2} \left(u - \frac{1}{2} \right)^2 - \frac{1}{32}. \quad (19)$$

Consequently

$$\mathfrak{W}(t) \geq 0, \quad -\frac{1}{32} \leq \Psi(t) \leq \frac{3}{32} \quad (t \geq 0). \quad (20)$$

Proposition 2 (Exact retained-remainder identity). *Put*

$$\mathcal{R}_{\text{dec}} := - \int_{(0,\tau)} \mathfrak{W}(t) dg(t), \quad (21)$$

$$\mathcal{R}_{\text{osc}} := \left(\Psi(T) + \frac{1}{32} \right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi(t) \right) dg(t). \quad (22)$$

Then both remainders are nonnegative, and

$$\begin{aligned} \mathcal{D}_{\text{rad}} &= \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau) \left(\frac{\tau}{4} + \frac{3}{32} \right) \\ &\quad + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}}. \end{aligned} \quad (23)$$

Therefore

$$W - P = \mathcal{B}(\rho, K) + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}} - \mathcal{E}_{\text{ang}}, \quad (24)$$

where

$$\mathcal{B}(\rho, K) := \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} + \frac{2\pi \rho K}{\arccos \rho} \left(\frac{K G_1(\rho)}{4} + \frac{3}{32} \right). \quad (25)$$

Proof. Distributionally, $dw = dt - dC$. Stieltjes integration by parts, using $F(T) = 0$ and $w(0) = 0$, gives

$$\mathcal{D}_{\text{rad}} = \int_0^T w(t) g(t) dt. \quad (26)$$

The improper boundary term at 0 vanishes because $\mathfrak{W}(t) = t^2/2$ there and $g(t) = O(t^{-1/3})$.

On $(0, \tau)$, integrate with $d\mathfrak{W} = w dt$:

$$\int_0^\tau w g = \mathfrak{W}(\tau) g(\tau) - \int_{(0,\tau)} \mathfrak{W} dg. \quad (27)$$

On (τ, T) , use $w = 1/4 + \Psi'$ and $\int_\tau^T g = F(\tau)$:

$$\int_\tau^T w g = \frac{F(\tau)}{4} + \Psi(T) g(T) - \Psi(\tau) g(\tau) - \int_{(\tau,T)} \Psi dg. \quad (28)$$

Since $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, adding (27) and (28) cancels the uncontrolled interface value $\Psi(\tau)$.

Now add and subtract the extremal sawtooth values in (20). The exact identity

$$\begin{aligned} &\Psi(T) g(T) - \int_{(\tau,T)} \Psi dg \\ &= -\frac{g(T)}{8} + \frac{3g(\tau)}{32} + \left(\Psi(T) + \frac{1}{32} \right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi \right) dg \end{aligned}$$

gives (23). By (8), $-dg$ is a positive measure on $(0, \tau)$ and dg is a positive measure on (τ, T) . Equations (20), (21), and (22) therefore prove nonnegativity of both remainders. Finally substitute (9) and $\tau = KG_1(\rho)$, and then use (16). $\square$

4 A smooth lower bound for the decreasing-branch remainder

The term $\mathcal{R}_{\text{dec}}$ is quantitatively important on the compact residual; discarding it loses too much. It has the following elementary smooth minorant.

Lemma 3 (A global primitive minorant). *For $t \geq 0$, define*

$$L(t) := \frac{t^2}{2(1+2t)}, \quad L'(t) = \frac{t(1+t)}{(1+2t)^2}. \quad (29)$$

Then

$$0 \leq L(t) \leq \mathfrak{W}(t) \quad (t \geq 0). \quad (30)$$

Proof. For $0 \leq t \leq 3/4$, one has $\mathfrak{W}(t) = t^2/2 \geq L(t)$. For $t \geq 3/4$, (20) gives

$$\mathfrak{W}(t) \geq \frac{t}{4} - \frac{1}{32}.$$

Moreover,

$$\frac{t}{4} - \frac{1}{32} - \frac{t^2}{2(1+2t)} = \frac{6t-1}{32(1+2t)} \geq 0.$$

This proves (30). $\square$

Define

$$\mathcal{R}_*(\rho, K) := - \int_{(0, \tau)} L(t) dg(t). \quad (31)$$

Because $-dg$ is positive on the decreasing branch,

$$0 \leq \mathcal{R}_* \leq \mathcal{R}_{\text{dec}}. \quad (32)$$

Integration by parts is legitimate at 0 because $L(t)g(t) = O(t^{5/3})$. Hence

$$\mathcal{R}_* = -L(\tau)g(\tau) + \int_0^\tau L'(t)g(t) dt. \quad (33)$$

On the outer branch $t = A(z)$, one has $g(t) dt = -2z dz$. Therefore

$$\mathcal{R}_* = -\frac{\tau^2}{2(1+2\tau)}g(\tau) + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz. \quad (34)$$

This expression is smooth in (ρ, K) on the compact residual and has no floor or wall discontinuity.

Corollary 4 (A concrete sufficient inequality). *Define*

$$\begin{aligned} \mathcal{H}(\rho, K) := & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi \rho K}{\arccos \rho} \left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32} \right) \\ & + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz, \quad \tau = KG_1(\rho). \end{aligned} \quad (35)$$

If

$$\mathcal{E}_{\text{ang}}(\rho, K) < \mathcal{H}(\rho, K), \quad (36)$$

then $P(\rho, K) < W(\rho, K)$.

Proof. By (25), (34), and

$$\frac{\tau}{4} - \frac{\tau^2}{2(1+2\tau)} = \frac{\tau}{4(1+2\tau)},$$

one has $\mathcal{H} = \mathcal{B} + \mathcal{R}_*$. Now use (24), (32), and $\mathcal{R}_{\text{osc}} \geq 0$. $\square$

5 The first unresolved inequality

The preceding calculation reduces the analytic replacement project to the following statement:

$$\boxed{\sum_{\substack{n \geq 1 \\ n-1/4 < T}} (M(R(n-1/4))^2 - R(n-1/4)^2) < \mathcal{H}(\rho, K)} \quad ((\rho, K) \in \mathcal{R}_c). \quad (37)$$

This is the first unresolved inequality in the present route. Everything to its right is the smooth action expression (35). Everything to its left is an explicitly centered angular rounding error. In particular, writing $r_n = m_n - 1/2 + \alpha_n$ gives its exact summands as (15); replacing them by $r_n + 1/4$ destroys the cancellation and is too coarse for the compact range.

The remaining discontinuities are transparent. For a fixed n , the summand is smooth until r_n crosses a half-integer. At $r_n = m + 1/2$, strict counting assigns $M(r_n) = m$; immediately on the side $r_n > m + 1/2$, the summand jumps upward by $2m + 1$. These are exactly the original proxy walls, now separated from the radial Stieltjes deficit.

There is a natural second summation problem behind (37). Put

$$e(r) := M(r)^2 - r^2, \quad q(r) := -A'(r), \quad h(t) := e(R(t)). \quad (38)$$

On every complete angular cell ($m \geq 1$),

$$\int_{m-1/2}^{m+1/2} (m^2 - r^2) dr = -\frac{1}{12}. \quad (39)$$

Thus the continuous comparison

$$\int_0^T h(t) dt = \int_0^K e(r) q(r) dr$$

has a favorable cell-average drift. Passing from this integral to the shifted samples $h(n - 1/4)$ produces the genuine double-sawtooth correlation. Its atoms contain the proxy-wall jumps and cannot simply be discarded by sign. In particular, if an angular wall coincides with a radial sample, Stieltjes integration by parts must retain the common-jump term dictated by the strict convention. A successful next lemma must pair those jump terms with the positive quantity already retained in $\mathcal{H}$, or bound them in blocks using the single-peaked shape of q from (6).

Remark 5 (Status). All identities and inequalities through Corollary 4 are analytic and require no finite computation. Equation (37) is a proposed replacement lemma, not a proved claim. Consequently this note does not by itself remove the compact certificate from the spherical-shell proof.

Angular Blocks for the Compact Shell Proxy

Round 2 analytic note

Abstract

The compact shell proxy can be reduced to an angular rounding defect $\mathcal{E}_{\text{ang}}$ and a smooth positive payment $\mathcal{H}$. This note attacks the remaining defect. First, an exact Stieltjes formula is given with the strict common-jump convention made explicit. The negative mean of every complete angular cell is combined with the single-peaked shape of $q = -A'$ to obtain continuous block estimates. A paired radial block argument then controls the full shifted sample sum, including all common jumps, by

$$\mathcal{E}_{\text{ang}} < \frac{(1 - \rho^2)K^2}{6} - \frac{N}{2},$$

where N is the number of active shifted radial layers. On the actual compact residual $N \geq 1$. The purely analytic replacement is therefore reduced to one smooth scalar inequality, displayed in (28). That scalar inequality is not proved here.

1 Definitions and the target payment

For $a > 0$ and $z \geq 0$, define

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z < a, \\ 0, & z \geq a. \end{cases} \quad (1)$$

Fix $0 < \rho < 1$ and $K > 0$, and put

$$A(z) := G_K(z) - G_{\rho K}(z), \quad T := A(0) = \frac{(1 - \rho)K}{\pi}. \quad (2)$$

The function A decreases strictly from T to 0. Denote its decreasing inverse by $R : [0, T] \rightarrow [0, K]$, and set

$$F(t) := R(t)^2, \quad q(z) := -A'(z), \quad \tau := A(\rho K) = KG_1(\rho). \quad (3)$$

Direct differentiation gives

$$q(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \frac{z}{K}, & \rho K \leq z < K. \end{cases} \quad (4)$$

Consequently

$$0 \leq q(z) \leq \frac{1}{2}, \quad q \text{ increases on } (0, \rho K), \quad q \text{ decreases on } (\rho K, K). \quad (5)$$

Let

$$x_n := n - \frac{1}{4}, \quad M(r) := \max \left\{ 0, \left\lceil r - \frac{1}{2} \right\rceil \right\}, \quad N := \#\{n \geq 1 : x_n < T\}. \quad (6)$$

For $1 \leq n \leq N$, write

$$r_n := R(x_n), \quad m_n := M(r_n), \quad (7)$$

and define the exact angular rounding defect

$$\mathcal{E}_{\text{ang}}(\rho, K) := \sum_{n=1}^N (m_n^2 - r_n^2). \quad (8)$$

The strict convention is built into M : at $r = m + 1/2$ one has $M(r) = m$, not $m + 1$.

The smooth payment obtained from the retained radial Stieltjes remainder is

$$\begin{aligned} \mathcal{H}(\rho, K) &:= \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ &\quad + \frac{2\pi \rho K}{\arccos \rho} \left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32} \right) \\ &\quad + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz. \end{aligned} \quad (9)$$

The preceding structural reduction proves

$$\mathcal{E}_{\text{ang}} < \mathcal{H} \implies P(\rho, K) < W(\rho, K). \quad (10)$$

This note derives a smooth upper bound for the left side.

2 Complete angular cells and the single peak

Define the angular sawtooth error

$$e(r) := M(r)^2 - r^2. \quad (11)$$

For $m \geq 1$, the strict m -cell is

$$I_m = (m - 1/2, m + 1/2], \quad e(r) = m^2 - r^2 \quad (r \in I_m).$$

Endpoint values do not affect the following Lebesgue integrals. The positive and negative half-cell masses are

$$P_m := \int_{m-1/2}^m (m^2 - r^2) dr = \frac{m}{4} - \frac{1}{24}, \quad (12)$$

$$Q_m := - \int_m^{m+1/2} (m^2 - r^2) dr = \frac{m}{4} + \frac{1}{24}. \quad (13)$$

In particular,

$$\int_{m-1/2}^{m+1/2} e(r) dr = P_m - Q_m = -\frac{1}{12}. \quad (14)$$

Lemma 1 (Weighted complete-cell bounds). *Let $m \geq 1$ and suppose the complete cell $[m - 1/2, m + 1/2]$ lies in $[0, K]$.*

1. If $m + 1/2 \leq \rho K$, then

$$\int_{m-1/2}^{m+1/2} e(r)q(r) dr \leq -\frac{q(m)}{12} \leq 0. \quad (15)$$

2. If $m - 1/2 \geq \rho K$, then

$$\int_{m-1/2}^{m+1/2} e(r)q(r) dr \leq P_m q(m - 1/2) - Q_m q(m + 1/2). \quad (16)$$

3. On the unique cell, if any, whose interior contains ρK ,

$$\int_{m-1/2}^{m+1/2} e(r)q(r) dr \leq P_m q(\rho K). \quad (17)$$

The initial partial cell $0 \leq r \leq 1/2$ contributes nonpositively.

Proof. On the left half of a cell $e \geq 0$, and on the right half $e \leq 0$. If q is increasing, then $q(r) \leq q(m)$ on the positive half and $q(r) \geq q(m)$ on the negative half. Equations (12)–(14) give (15).

If q is decreasing, then $q(r) \leq q(m - 1/2)$ on the positive half and $q(r) \geq q(m + 1/2)$ on the negative half. This gives (16). On the peak cell, discard the negative half and use $q \leq q(\rho K)$ on the positive half. Finally, $M(r) = 0$ and $e(r) = -r^2$ for $0 \leq r \leq 1/2$. $\square$

Thus the continuous angular defect has a favorable analytic block structure:

$$\int_0^T e(R(t)) dt = \int_0^K e(r)q(r) dr. \quad (18)$$

All complete cells below the peak are nonpositive, and the decreasing side is controlled by the endpoint differences in (16). The difficulty is passing from the integral in (18) to the shifted samples in (8); the next subsection records that passage with all common jumps included.

3 Exact common-jump convention

Let

$$h(t) := e(R(t)), \quad C(t) := \#\{n : 0 < x_n \leq t, x_n < T\}, \quad p(t) := C(t) - t. \quad (19)$$

Choose the right-continuous versions of C, p , and h . Right continuity of h is exactly the strict angular convention: when $R(t) = m + 1/2$, the post-jump value uses $M = m$. The atomic measure dC therefore samples the correct value even when a radial point x_n is also an angular wall.

Lemma 2 (Euler identity with common jumps). *With the preceding representatives,*

$$\mathcal{E}_{\text{ang}} = \int_0^T h(t) dt - \int_{(0,T]} p(t^-) dh(t). \quad (20)$$

At a common radial-angular wall the second integral uses the left value $p(t^-) = -3/4$; no extra or omitted wall term is permitted.

Proof. The measure dC has a unit atom at every active x_n , so

$$\mathcal{E}_{\text{ang}} = \int_{(0,T)} h(t) dC(t).$$

Since $dC = dt + dp$,

$$\mathcal{E}_{\text{ang}} = \int_0^T h(t) dt + \int_{(0,T]} h(t) dp(t).$$

For right-continuous functions of bounded variation, the product formula in the form

$$\int_{(0,T]} h(t) dp(t) = [hp]_0^T - \int_{(0,T]} p(t^-) dh(t)$$

already contains the product of simultaneous jumps. Here $p(0) = 0$ and $h(T) = e(0) = 0$, so the boundary term vanishes. This proves (20). If $t = x_n$, then $C(t^-) = n - 1$ and hence $p(t^-) = n - 1 - (n - 1/4) = -3/4$, including a common angular wall. $\square$

Equation (20) explains why the negative means in (14) cannot be used without a jump audit. The second term contains both the continuous rise inside each angular cell and the downward atom at its strict lower wall. A direct sign deletion would be invalid. The paired estimate below controls the integral and all its atoms simultaneously.

4 A paired radial-block estimate

The cap $q \leq 1/2$ in (5) forces a useful separation: one unit of radial action spans at least two units of angular radius. The shift $x_n = n - 1/4$ leaves a left block of length $3/4$, enough to pay the worst angular rounding in that layer.

Lemma 3 (One-layer paired block). *For every active n ,*

$$r_n \leq \frac{4}{3} \int_{n-1}^{x_n} R(t) dt - \frac{3}{4}, \quad (21)$$

and

$$m_n^2 - r_n^2 < r_n + \frac{1}{4}. \quad (22)$$

Both statements remain valid, with the displayed strictness, at every radial-angular common wall.

Proof. Fix $t \in [n - 1, x_n]$ and put $z = R(t) \geq r_n$. Since $A(r_n) = x_n$, (5) gives

$$x_n - t = A(r_n) - A(z) = \int_{r_n}^z q(s) ds \leq \frac{z - r_n}{2}.$$

Hence

$$R(t) \geq r_n + 2(x_n - t).$$

Integration over the interval of length $3/4$ yields

$$\int_{n-1}^{x_n} R(t) dt \geq \frac{3}{4} r_n + \frac{9}{16},$$

which is (21).

By strict counting, $m_n < r_n + 1/2$ even when r_n is a half-integer: at $r_n = m + 1/2$ the value is $m_n = m$. Squaring gives

$$m_n^2 < (r_n + 1/2)^2 = r_n^2 + r_n + 1/4,$$

and proves (22). $\square$

Proposition 4 (Global angular block bound). *For every $0 < \rho < 1$ and every $K > 0$ with $N \geq 1$,*

$$\boxed{\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1 - \rho^2)K^2}{6} - \frac{N}{2}}. \quad (23)$$

In particular,

$$\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1 - \rho^2)K^2}{6} - \frac{1}{2}. \quad (24)$$

Proof. The left blocks $[n - 1, x_n]$, $1 \leq n \leq N$, are disjoint subsets of $[0, T]$. Summing (21) and then (22) gives

$$\begin{aligned} \mathcal{E}_{\text{ang}} &< \sum_{n=1}^N \left(r_n + \frac{1}{4} \right) \\ &\leq \frac{4}{3} \sum_{n=1}^N \int_{n-1}^{x_n} R(t) dt - \frac{N}{2} \\ &\leq \frac{4}{3} \int_0^T R(t) dt - \frac{N}{2}. \end{aligned}$$

The area identity for decreasing inverses gives

$$\int_0^T R(t) dt = \int_0^K A(z) dz.$$

Scaling $z = as$ in (1) and using

$$\int_0^1 \sqrt{1 - s^2} ds = \frac{\pi}{4}, \quad \int_0^1 s \arccos s ds = \frac{\pi}{8},$$

shows that

$$\int_0^a G_a(z) dz = \frac{a^2}{8}.$$

Consequently

$$\int_0^T R(t) dt = \frac{(1 - \rho^2)K^2}{8},$$

which proves (23). Equation (24) follows from $N \geq 1$. □

5 The remaining scalar inequality

Set

$$\rho_c := \frac{1}{1 + 2\pi}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1 - \rho)^2} + 132}, \quad (25)$$

and consider the closed superset of the actual compact residual

$$\mathcal{R}_c := \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad k_{11}(\rho) \leq K \leq 200 \right\}. \quad (26)$$

On this set,

$$T \geq \frac{(1 - \rho)k_{11}(\rho)}{\pi} = \sqrt{1 + \frac{132(1 - \rho)^2}{\pi^2}} > 1, \quad (27)$$

so $N \geq 1$. Proposition 4 and (10) reduce the analytic compact closure to the single inequality

$$\boxed{\mathcal{S}(\rho, K) := \mathcal{H}(\rho, K) - \frac{(1 - \rho^2)K^2}{6} + \frac{1}{2} > 0} \quad ((\rho, K) \in \mathcal{R}_c). \quad (28)$$

Everything in $\mathcal{S}$ is smooth on a neighborhood of $\mathcal{R}_c$; the floor functions and all radial-angular common jumps have disappeared. Substituting (9) gives the explicit form

$$\begin{aligned} \mathcal{S}(\rho, K) = & \frac{(5\rho^2 - 2)K^2}{12} - \frac{\pi\rho K}{4(1 - \rho)} + \frac{1}{2} \\ & + \frac{2\pi\rho K}{\arccos \rho} \left(\frac{KG_1(\rho)}{4(1 + 2KG_1(\rho))} + \frac{3}{32} \right) \\ & + \int_{\rho K}^K 2z \frac{A(z)(1 + A(z))}{(1 + 2A(z))^2} dz. \end{aligned} \quad (29)$$

The elementary identity

$$\frac{u(1 + u)}{(1 + 2u)^2} = \frac{1}{4} - \frac{1}{4(1 + 2u)^2} \quad (30)$$

turns the same scalar gap into the more useful loss form

$$\begin{aligned} \mathcal{S}(\rho, K) = & \frac{(1 + 2\rho^2)K^2}{12} - \frac{1}{2} \int_{\rho K}^K \frac{z}{(1 + 2A(z))^2} dz \\ & - \frac{\pi\rho K}{4(1 - \rho)} + \frac{1}{2} + \frac{2\pi\rho K}{\arccos \rho} \left(\frac{KG_1(\rho)}{4(1 + 2KG_1(\rho))} + \frac{3}{32} \right). \end{aligned} \quad (31)$$

Thus the only non-endpoint quantity still requiring control is the positive boundary-layer loss integral in (31).

Remark 5 (Status and next step). The complete-cell bounds, the common-jump formula, and the global estimate (23) are proved analytically. The first remaining step in this route is the smooth scalar inequality (28). Proving it will require a uniform lower bound for the positive outer-action integral in (29), or, equivalently, a uniform upper bound for the boundary-layer loss in (31), together with the explicit endpoint terms. Until that inequality is established, this note does not replace the compact interval certificate.

Purely Analytic Positivity of the Compact Scalar Gap

Round 3 analytic note

Abstract

The angular-block reduction for the compact spherical-shell proxy leaves one smooth scalar inequality. This note proves that inequality without interval arithmetic or a finite numerical certificate. The loss integral is first differentiated with respect to the spectral cutoff. A turning-point lower bound and an exact beta integral show that the scalar gap is strictly increasing in that cutoff. At its lower endpoint, a rational secant majorant gives a uniform analytic estimate for the loss. The remaining one-variable inequality is then reduced to positivity of a degree-nine polynomial; all of its Bernstein coefficients on one rational interval are displayed and are positive. Thus the compact scalar gap is positive throughout the stated residual region.

1 The scalar gap

For $0 \leq s \leq 1$, put

$$G_1(s) := \frac{1}{\pi} \left(\sqrt{1-s^2} - s \arccos s \right). \quad (1)$$

Fix

$$\rho_c := \frac{1}{1+2\pi}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}, \quad (2)$$

and consider

$$\mathcal{R}_c := \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad k_{11}(\rho) \leq K \leq 200 \right\}. \quad (3)$$

The scalar gap left by the angular-block estimate is

$$\begin{aligned} \mathcal{S}(\rho, K) := & \frac{(1+2\rho^2)K^2}{12} - \mathcal{J}(\rho, K) - \frac{\pi\rho K}{4(1-\rho)} + \frac{1}{2} \\ & + \frac{2\pi\rho K}{\arccos \rho} \left(\frac{KG_1(\rho)}{4(1+2KG_1(\rho))} + \frac{3}{32} \right), \end{aligned} \quad (4)$$

where

$$\mathcal{J}(\rho, K) := \frac{K^2}{2} \int_{\rho}^1 \frac{s}{(1+2KG_1(s))^2} ds. \quad (5)$$

Theorem 1. *For every $(\rho, K) \in \mathcal{R}_c$,*

$$\boxed{\mathcal{S}(\rho, K) > 0.} \quad (6)$$

We use only the elementary rational estimates

$$\frac{157}{50} < \pi < \frac{22}{7}, \quad \frac{140}{99} < \sqrt{2}, \quad \frac{5}{3} < \sqrt{3}. \quad (7)$$

The two square-root estimates follow immediately by squaring; the displayed bounds for π are the classical Archimedean bounds (with the lower one weakened to $157/50$).

2 Strict monotonicity in the spectral cutoff

The identity

$$G_1(s) = \frac{1}{\pi} \int_s^1 \arccos u \, du \quad (8)$$

and $1 - \cos v \leq v^2/2$ imply

$$G_1(s) \geq \frac{2\sqrt{2}}{3\pi} (1-s)^{3/2}. \quad (9)$$

Indeed, $\arccos u \geq \sqrt{2(1-u)}$, and integration gives (9).

Lemma 2 (Derivative of the boundary-layer loss). *For fixed ρ ,*

$$\partial_K \mathcal{J}(\rho, K) = K \int_\rho^1 \frac{s}{(1 + 2KG_1(s))^3} \, ds. \quad (10)$$

Moreover, if $K > 12$, then

$$\int_\rho^1 \frac{s}{(1 + 2KG_1(s))^3} \, ds < \frac{88}{567}. \quad (11)$$

Proof. Differentiating (5) under the integral sign gives

$$\begin{aligned} \partial_K \mathcal{J} &= K \int_\rho^1 \frac{s}{(1 + 2KG_1(s))^2} \, ds - 2K^2 \int_\rho^1 \frac{sG_1(s)}{(1 + 2KG_1(s))^3} \, ds \\ &= K \int_\rho^1 \frac{s}{(1 + 2KG_1(s))^3} \, ds, \end{aligned}$$

which proves (10).

Set $a = 4\sqrt{2}/(3\pi)$. By (9), the change of variable $t = 1 - s$, and $s \leq 1$,

$$\begin{aligned} \int_\rho^1 \frac{s \, ds}{(1 + 2KG_1(s))^3} &\leq \int_0^\infty \frac{dt}{(1 + aKt^{3/2})^3} \\ &= \frac{8\pi}{27\sqrt{3}} (aK)^{-2/3}. \end{aligned} \quad (12)$$

The last equality is the beta integral

$$\frac{2}{3} (aK)^{-2/3} B\left(\frac{2}{3}, \frac{7}{3}\right) = \frac{8\pi}{27\sqrt{3}} (aK)^{-2/3}.$$

At $K = 12$, the cube of the last member of (12) is

$$\frac{\pi^5}{59049\sqrt{3}}.$$

Using (7),

$$\frac{\pi^5}{59049\sqrt{3}} < \frac{5153632}{1654060905} < \left(\frac{88}{567}\right)^3,$$

where the second strict inequality is the exact identity

$$\left(\frac{88}{567}\right)^3 - \frac{5153632}{1654060905} = \frac{27812576}{44659644435} > 0.$$

The right side of (12) decreases with K , proving (11). □

Proposition 3 (Monotonicity of the scalar gap). *For every $(\rho, K) \in \mathcal{R}_c$,*

$$\partial_K \mathcal{S}(\rho, K) > 0. \quad (13)$$

Consequently,

$$\mathcal{S}(\rho, K) \geq \mathcal{S}(\rho, k_{11}(\rho)). \quad (14)$$

Proof. Write

$$\theta := \arccos \rho, \quad y := KG_1(\rho).$$

The derivative of the final line of (4) is

$$\frac{2\pi\rho}{\theta} \left(\frac{y(1+y)}{2(1+2y)^2} + \frac{3}{32} \right) \geq \frac{3\rho}{8}, \quad (15)$$

because $\theta \leq \pi/2$. Also

$$-\frac{\pi\rho}{4(1-\rho)} > -\frac{11\rho}{14(1-\rho)}.$$

Every K in $\mathcal{R}_c$ is greater than 12; this also follows from the stronger estimate established in Lemma 4 below. Lemma 2 therefore gives

$$\begin{aligned} \partial_K \mathcal{S} &> K \left(\frac{1+2\rho^2}{6} - \frac{88}{567} \right) + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)} \\ &= K \left(\frac{13}{1134} + \frac{\rho^2}{3} \right) + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)} \\ &> \frac{26}{189} + 4\rho^2 + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)}. \end{aligned} \quad (16)$$

It remains to check the sign of the last expression. Multiplication by $1-\rho$ produces

$$p(\rho) := (1-\rho) \left(\frac{26}{189} + 4\rho^2 + \frac{3\rho}{8} \right) - \frac{11\rho}{14}. \quad (17)$$

The bound $\pi < 22/7$ gives $\rho_c > 7/51$. On the larger rational interval $[7/51, 39/50]$, set

$$u := \frac{\rho - 7/51}{39/50 - 7/51} \in [0, 1].$$

In the degree-three Bernstein basis $B_{j,3}(u) = \binom{3}{j} u^j (1-u)^{3-j}$, exact expansion gives

$$p(\rho) = \sum_{j=0}^3 b_j B_{j,3}(u), \quad (18)$$

with

j	b_j
0	$\frac{335005}{32947387}$
1	$\frac{196635600}{281783183}$
2	$\frac{578340000}{231517}$
3	$\frac{13500000}{13500000}$

(19)

All four coefficients are positive. Since the Bernstein basis functions are nonnegative and sum to one, $p(\rho) > 0$. Equation (16) proves (13), and (14) follows by integration in K . $\square$

3 An exact rational upper bound for the loss

Lemma 4 (Uniform loss bound). *For every $(\rho, K) \in \mathcal{R}_c$,*

$$K > \frac{241}{20} \quad (20)$$

and

$$\int_{\rho}^1 \frac{x}{(1 + 2KG_1(x))^2} dx < \frac{17}{100}. \quad (21)$$

Proof. Since $\rho \geq \rho_c$,

$$\frac{\pi}{1 - \rho} \geq \frac{\pi}{1 - \rho_c} = \pi + \frac{1}{2} > \frac{91}{25}.$$

Consequently

$$K^2 \geq k_{11}(\rho)^2 > 132 + \left(\frac{91}{25}\right)^2 = \left(\frac{241}{20}\right)^2 + \frac{471}{10000},$$

which proves (20). The rational bounds in (7) now give

$$\frac{4\sqrt{2}K}{3\pi} > \frac{4(140/99)(241/20)}{3(22/7)} = \frac{23618}{3267} = \frac{36}{5} + \frac{478}{16335}. \quad (22)$$

Put $y = 1 - x$. Equations (9) and (22) yield

$$\begin{aligned} \int_{\rho}^1 \frac{x dx}{(1 + 2KG_1(x))^2} &< \int_0^{1-\rho} \frac{1-y}{(1 + (36/5)y^{3/2})^2} dy \\ &\leq \int_0^1 \frac{1-y}{(1 + (36/5)y^{3/2})^2} dy. \end{aligned} \quad (23)$$

We bound the last integral by exact rational arithmetic. Set

$$c := \frac{36}{5}, \quad \phi(z) := \frac{1}{(1+z)^2},$$

and use the rational grid

$$(t_i)_{i=0}^{10} = \left(0, \frac{1}{4}, \frac{5}{16}, \frac{3}{8}, \frac{7}{16}, \frac{1}{2}, \frac{9}{16}, \frac{5}{8}, \frac{3}{4}, \frac{7}{8}, 1\right). \quad (24)$$

For $z_i = ct_i^3$, let

$$\beta_i := \frac{\phi(z_{i+1}) - \phi(z_i)}{z_{i+1} - z_i}, \quad \alpha_i := \phi(z_i) - \beta_i z_i.$$

Because $\phi''(z) = 6(1+z)^{-4} > 0$, its secant satisfies

$$\phi(z) \leq \alpha_i + \beta_i z \quad (z_i \leq z \leq z_{i+1}).$$

After $y = t^2$, the integral in (23) is

$$\int_0^1 2t(1-t^2)\phi(ct^3) dt.$$

Thus it is at most

$$Q := \sum_{i=0}^9 (F_i(t_{i+1}) - F_i(t_i)), \quad (25)$$

where the exact polynomial antiderivative is

$$F_i(t) := \alpha_i \left(t^2 - \frac{t^4}{2} \right) + 2c\beta_i \left(\frac{t^5}{5} - \frac{t^7}{7} \right). \quad (26)$$

Every number in (25) is rational. Substitution of the grid (24) gives

$$Q = \frac{1305767238061446154064110434835938779253983373178558326887}{7706446260271137290663804190970969288616921855152043745280}, \quad (27)$$

and direct subtraction gives

$$\frac{17}{100} - Q = \frac{21643130923235926743681388145629999054466710986445549053}{38532231301355686453319020954854846443084609275760218726400} > 0. \quad (28)$$

Equations (23)–(28) prove (21). $\square$

4 The base curve

Fix ρ and now set

$$t := \sqrt{1 - \rho}, \quad k := k_{11}(\rho) = \sqrt{132 + \frac{\pi^2}{t^4}}, \quad \rho = 1 - t^2. \quad (29)$$

The compact range lies strictly inside the rational interval

$$\frac{7}{15} < t < \frac{14}{15}. \quad (30)$$

Indeed,

$$t^2 \geq \frac{11}{50} > \left(\frac{7}{15} \right)^2,$$

while $\rho \geq \rho_c > 7/51$ gives

$$t^2 = 1 - \rho < \frac{44}{51} < \left(\frac{14}{15} \right)^2.$$

Lemma 5 (Analytic endpoint payment). *Let*

$$C(\rho) := \frac{100\rho^2 - 1}{600} \quad (31)$$

and

$$E(t) := \frac{11}{14t^2} - \frac{14}{15(1+2t)} - \frac{3}{8t}. \quad (32)$$

Then $C(\rho) > 0$, $E(t) > 0$, and

$$\mathcal{S}(\rho, k) > C(\rho)k^2 - \rho E(t)k + \frac{1}{2}. \quad (33)$$

Proof. First, $\rho \geq \rho_c > 7/51 > 1/10$, so $C(\rho) > 0$. Lemma 4 and

$$\frac{1 + 2\rho^2}{12} - \frac{17}{200} = \frac{100\rho^2 - 1}{600}$$

show that the first two terms in (4) are strictly greater than $C(\rho)k^2$.

Set $\theta = \arccos \rho$ and $Y = kG_1(\rho)$. For $0 \leq t \leq 1$,

$$\theta = \arccos(1 - t^2) = 2 \arcsin \frac{t}{\sqrt{2}} \leq \frac{\pi t}{2}. \quad (34)$$

To see the last inequality, use convexity of $\arcsin$ on $[0, 1/\sqrt{2}]$: its graph lies below the chord joining $(0, 0)$ to $(1/\sqrt{2}, \pi/4)$. Furthermore, (9) and $k > \pi/t^2$ give

$$Y > \frac{2\sqrt{2}}{3}t > \frac{14}{15}t. \quad (35)$$

Since $Y \mapsto Y/(1 + 2Y)$ is increasing, (34)–(35) imply

$$\begin{aligned} \frac{2\pi\rho k}{\theta} \left(\frac{Y}{4(1 + 2Y)} + \frac{3}{32} \right) \\ > \rho k \left(\frac{14}{15(1 + 2t)} + \frac{3}{8t} \right). \end{aligned} \quad (36)$$

Also

$$-\frac{\pi\rho k}{4t^2} > -\frac{11\rho k}{14t^2}.$$

Combining these inequalities with (4) proves (33).

Finally,

$$840t^2(1 + 2t)E(t) = 660 + 1005t - 1414t^2. \quad (37)$$

The quadratic on the right is concave, so its minimum on $[7/15, 14/15]$ occurs at an endpoint. Its endpoint values are

$$\frac{184739}{225}, \quad \frac{82406}{225},$$

respectively. Both are positive, proving $E(t) > 0$. $\square$

Proposition 6 (Rational positivity on the base curve). *For every $\rho_c \leq \rho \leq 39/50$,*

$$\mathcal{S}(\rho, k_{11}(\rho)) > 0. \quad (38)$$

Proof. The elementary inequality

$$\sqrt{x^2 + 132} < x + \frac{66}{x} \quad (x > 0)$$

and (7) give

$$k^2 > 132 + \frac{(157/50)^2}{t^4}, \quad (39)$$

$$k < \frac{22}{7t^2} + \frac{3300t^2}{157}. \quad (40)$$

Since $C(\rho) > 0$ and $\rho E(t) > 0$, Lemma 5 yields

$$\mathcal{S}(\rho, k) > R(t), \quad (41)$$

where

$$\begin{aligned} R(t) := & C(1 - t^2) \left(132 + \frac{(157/50)^2}{t^4} \right) \\ & - (1 - t^2)E(t) \left(\frac{22}{7t^2} + \frac{3300t^2}{157} \right) + \frac{1}{2}. \end{aligned} \quad (42)$$

It remains only exact polynomial algebra. Put

$$H(t) := t^4(1 + 2t)R(t).$$

Using (37), its factorized form is

$$\begin{aligned} H(t) = & C(1 - t^2)(1 + 2t) \left(132t^4 + \left(\frac{157}{50} \right)^2 \right) \\ & - \frac{(1 - t^2)(660 + 1005t - 1414t^2)}{840} \left(\frac{22}{7} + \frac{3300}{157}t^4 \right) \\ & + \frac{1}{2}t^4(1 + 2t). \end{aligned} \quad (43)$$

For an independently checkable expansion,

$$\begin{aligned} H(t) = & -\frac{20642567}{24500000} - \frac{6205067}{12250000}t + \frac{547983}{122500}t^2 - \frac{1033727}{367500}t^3 \\ & + \frac{34911551}{16485000}t^4 + \frac{187093801}{8242500}t^5 + \frac{8679}{1099}t^6 \\ & - \frac{138149}{2198}t^7 - \frac{2101}{157}t^8 + 44t^9. \end{aligned} \quad (44)$$

Set

$$u := \frac{t - 7/15}{7/15} \in [0, 1].$$

In the degree-nine Bernstein basis,

$$H(t) = \sum_{j=0}^9 d_j B_{j,9}(u), \quad B_{j,9}(u) = \binom{9}{j} u^j (1 - u)^{9-j}, \quad (45)$$

where exact conversion of (44) gives

j	d_j
0	$\frac{65408801807437}{9463832437500000}$
1	$\frac{299549381760793}{1135659892500000}$
2	$\frac{15904699805720773}{28391497312500000}$
3	$\frac{2765605256590021}{3154610812500000}$
4	$\frac{33203943051258761}{28391497312500000}$
5	$\frac{38636221171331747}{28391497312500000}$
6	$\frac{2538003206646073}{1892766487500000}$
7	$\frac{28758289199580211}{28391497312500000}$
8	$\frac{12920786461478803}{28391497312500000}$
9	$\frac{3428732855971679}{9463832437500000}$

(46)

Every d_j is positive. The Bernstein basis functions are nonnegative and sum to one; hence $H(t) > 0$ throughout $[7/15, 14/15]$. Since $t^4(1+2t) > 0$, this proves $R(t) > 0$. Equation (41) proves (38). $\square$

Proof of Theorem 1. Proposition 3 reduces every point of $\mathcal{R}_c$ to the base curve $K = k_{11}(\rho)$. Proposition 6 proves strict positivity on that curve. Therefore $\mathcal{S}(\rho, K) > 0$ throughout $\mathcal{R}_c$. $\square$

Remark 7 (Nature of the certificate). The rational secants and Bernstein expansions above are finite exact algebra, not interval arithmetic: convexity proves the functional majorant on each secant interval, and positivity follows from explicitly displayed rational coefficients. No floating-point evaluation, sampled minimum, or numerical root isolation is used as a premise.

A Hand-Checkable Proof of Positivity for the Aggregate Curvature Floor

Abstract

We prove, without numerical computation, the strict positivity of the one-variable function that supplies the curvature floor in the aggregate-tail argument for spherical shells. Every estimate is reduced to an exact rational comparison. In fact, on the full required interval we obtain the uniform lower bound

$$F(\rho) > \frac{1549}{84150} > 0.$$

1 Definitions and elementary estimates

For $0 \leq s \leq 1$, define

$$G_1(s) := \frac{1}{\pi} \left(\sqrt{1-s^2} - s \arccos s \right). \quad (1)$$

For $0 < \rho < 1$, put

$$\eta_\rho := G_1(\max\{\rho, 1/2\}), \quad d_\rho := \frac{2 \arcsin \rho}{\pi}, \quad b_\rho := \frac{2\pi\rho}{1-\rho}. \quad (2)$$

The function to be estimated is

$$F(\rho) := 2\rho\eta_\rho + d_\rho\rho^2 - \frac{3(1+d_\rho)b_\rho}{800}. \quad (3)$$

The interval in question is well inside $(0, 1)$, since

$$0 < \frac{7}{51} < \frac{1}{2} < \frac{39}{50} < 1;$$

the two nontrivial middle comparisons are respectively $14 < 51$ and $25 < 39$. Thus all quantities in (2) and (3) are well defined there.

We first record all numerical inequalities used below, together with exact proofs.

Lemma 1 (Rational bounds for π and square roots). *One has*

$$\pi < \frac{22}{7}, \quad \sqrt{2} > \frac{7}{5}, \quad \sqrt{3} > \frac{5}{3}, \quad \sqrt{\frac{11}{50}} > \frac{7}{15}. \quad (4)$$

Proof. Polynomial division gives the identity

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

The left-hand side is positive for $0 < x < 1$. Hence

$$\begin{aligned} 0 &< \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx \\ &= \frac{1}{7} - \frac{4}{6} + 1 - \frac{4}{3} + 4 - 4 \arctan(1) \\ &= \frac{22}{7} - \pi, \end{aligned}$$

where $\arctan(1) = \pi/4$. This proves $\pi < 22/7$.

All three square-root estimates follow by squaring positive quantities:

$$2 - \left(\frac{7}{5}\right)^2 = \frac{1}{25} > 0, \quad 3 - \left(\frac{5}{3}\right)^2 = \frac{2}{9} > 0,$$

and

$$\frac{11}{50} - \left(\frac{7}{15}\right)^2 = \frac{11}{50} - \frac{49}{225} = \frac{1}{450} > 0.$$

□

Lemma 2 (Integral representation and a lower bound for G_1). *For $0 \leq s \leq 1$,*

$$G_1(s) = \frac{1}{\pi} \int_s^1 \arccos t \, dt \geq \frac{2\sqrt{2}}{3\pi} (1-s)^{3/2}. \quad (5)$$

In particular, if $1/2 \leq s \leq 39/50$, then

$$G_1(s) > \frac{343}{11250}. \quad (6)$$

Proof. The antiderivative

$$\int \arccos t \, dt = t \arccos t - \sqrt{1-t^2}$$

shows directly that

$$\int_s^1 \arccos t \, dt = \sqrt{1-s^2} - s \arccos s,$$

which proves the first equality in (5).

For $0 \leq t \leq 1$, let $y = \arccos t$, so $0 \leq y \leq \pi/2$. Since $\sin u \leq u$ for $u \geq 0$,

$$\sqrt{2(1-t)} = \sqrt{2(1-\cos y)} = 2 \sin \frac{y}{2} \leq y = \arccos t.$$

Integrating this pointwise inequality gives

$$\begin{aligned} G_1(s) &\geq \frac{1}{\pi} \int_s^1 \sqrt{2(1-t)} \, dt \\ &= \frac{2\sqrt{2}}{3\pi} (1-s)^{3/2}, \end{aligned}$$

which completes (5). For completeness, $\sin u \leq u$ follows because the derivative of $u - \sin u$ is $1 - \cos u \geq 0$, and the function vanishes at zero.

Now assume $1/2 \leq s \leq 39/50$. Then $1 - s \geq 11/50$. Lemma 1 yields

$$\begin{aligned} G_1(s) &\geq \frac{2\sqrt{2}}{3\pi}(1-s)^{3/2} \\ &\geq \frac{2\sqrt{2}}{3\pi} \left(\frac{11}{50}\right)^{3/2} \\ &> \frac{2(7/5)}{3(22/7)} \frac{11}{50} \frac{7}{15} \\ &= \frac{49}{165} \frac{77}{750} = \frac{343}{11250}. \end{aligned}$$

Here the last equality follows by cancelling a factor 11 from 3773/123750. $\square$

2 Strict positivity of the curvature floor

Theorem 3. *For every*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50},$$

the function in (3) satisfies

$$F(\rho) > \frac{1549}{84150} > 0. \quad (7)$$

Proof. Because $\rho > 0$, substitution of $b_\rho = 2\pi\rho/(1-\rho)$ into (3) and division by ρ give the exact identity

$$\frac{F(\rho)}{\rho} = 2\eta_\rho + d_\rho\rho - \frac{3\pi(1+d_\rho)}{400(1-\rho)}. \quad (8)$$

We treat the two branches in the definition of η_ρ separately.

First branch: $7/51 \leq \rho \leq 1/2$. Here $\max\{\rho, 1/2\} = 1/2$, and therefore

$$\begin{aligned} \eta_\rho &= G_1(1/2) = \frac{1}{\pi} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \arccos \frac{1}{2} \right) \\ &= \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \end{aligned} \quad (9)$$

because $\arccos(1/2) = \pi/3$. Since $\arcsin$ is increasing on $[0, 1]$,

$$0 < d_\rho = \frac{2 \arcsin \rho}{\pi} \leq \frac{2 \arcsin(1/2)}{\pi} = \frac{1}{3}.$$

Also $1 - \rho \geq 1/2$, so $(1 - \rho)^{-1} \leq 2$. It follows that

$$\frac{3\pi(1+d_\rho)}{400(1-\rho)} \leq \frac{3\pi(4/3)2}{400} = \frac{\pi}{50} < \frac{11}{175}. \quad (10)$$

On the other hand, Lemma 1 gives

$$2\eta_\rho = \frac{\sqrt{3}}{\pi} - \frac{1}{3} > \frac{5/3}{22/7} - \frac{1}{3} = \frac{35}{66} - \frac{22}{66} = \frac{13}{66}. \quad (11)$$

The middle term $d_\rho \rho$ in (8) is nonnegative. Combining (8), (10), and (11), we obtain

$$\frac{F(\rho)}{\rho} > \frac{13}{66} - \frac{11}{175} = \frac{2275 - 726}{11550} = \frac{1549}{11550}. \quad (12)$$

Since $\rho \geq 7/51$ and $11550 = 7 \cdot 1650$, this implies

$$F(\rho) > \frac{7}{51} \frac{1549}{11550} = \frac{1549}{84150}. \quad (13)$$

Second branch: $1/2 \leq \rho \leq 39/50$. Here $\max\{\rho, 1/2\} = \rho$, so $\eta_\rho = G_1(\rho)$. Lemma 2 gives

$$2\eta_\rho > \frac{343}{5625}. \quad (14)$$

Monotonicity of $\arcsin$, together with $\rho \geq 1/2$, gives

$$d_\rho \geq \frac{1}{3}, \quad d_\rho \rho \geq \frac{1}{6}. \quad (15)$$

For the required upper bound on d_ρ , observe that

$$\left(\frac{39}{50}\right)^2 = \frac{1521}{2500} < \frac{1875}{2500} = \frac{3}{4} = \left(\frac{\sqrt{3}}{2}\right)^2.$$

Both sides are positive, hence $39/50 < \sqrt{3}/2$. Consequently

$$d_\rho \leq \frac{2 \arcsin(39/50)}{\pi} < \frac{2 \arcsin(\sqrt{3}/2)}{\pi} = \frac{2}{3}. \quad (16)$$

Finally, $1 - \rho \geq 11/50$, so $(1 - \rho)^{-1} \leq 50/11$. From (16) and Lemma 1,

$$\begin{aligned} \frac{3\pi(1 + d_\rho)}{400(1 - \rho)} &< \frac{3\pi(5/3)}{400} \frac{50}{11} = \frac{5\pi}{88} \\ &< \frac{5(22/7)}{88} = \frac{5}{28}. \end{aligned} \quad (17)$$

Substitution of (14), (15), and (17) into (8) yields

$$\begin{aligned} \frac{F(\rho)}{\rho} &> \frac{343}{5625} + \frac{1}{6} - \frac{5}{28} \\ &= \frac{9604 + 26250 - 28125}{157500} \\ &= \frac{7729}{157500}. \end{aligned} \quad (18)$$

Since $\rho \geq 1/2$,

$$F(\rho) > \frac{7729}{315000}. \quad (19)$$

The high-branch lower bound is stronger than the one asserted in (7), because exact subtraction gives

$$\frac{7729}{315000} - \frac{1549}{84150} = \frac{120341}{19635000} > 0.$$

The two branches cover the full interval. Their definitions agree at $\rho = 1/2$, because

$$G_1(\max\{1/2, 1/2\}) = G_1(1/2).$$

Consequently, (13) and (19) prove (7) everywhere on $[7/51, 39/50]$. $\square$

Analytic Positivity of the Aggregate Base Derivative

For $0 \leq s \leq 1$, set

$$G_1(s) = \frac{\sqrt{1-s^2} - s \arccos s}{\pi}.$$

For $0 < \rho < 1$ and $K > 0$, put

$$\eta_\rho = G_1(\max\{\rho, 1/2\}), \quad d_\rho = \frac{2 \arcsin \rho}{\pi}, \quad b_\rho = \frac{2\pi\rho}{1-\rho},$$

$$\mu = \rho K, \quad \bar{R} = \mu + \frac{1}{2}, \quad S = \sqrt{\bar{R}^2 + b_\rho K},$$

$$I(\rho, K) = \frac{1}{2} \left[\bar{R}S + b_\rho K \operatorname{arsinh} \frac{\bar{R}}{\sqrt{b_\rho K}} - \bar{R}^2 \right],$$

and define the continuous aggregate reserve

$$\begin{aligned} \mathcal{B}(\rho, K) &= (\mu - \tfrac{1}{2})(K\eta_\rho - 1) + \frac{d_\rho}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) \\ &\quad - (1 + d_\rho)I(\rho, K) - \frac{8(\mu + \frac{1}{2})}{15\sqrt{\mu}}. \end{aligned} \tag{1}$$

Direct differentiation gives

$$I_K = \rho(S - \bar{R}) + \frac{b_\rho}{2} \operatorname{arsinh} \frac{\bar{R}}{\sqrt{b_\rho K}}, \tag{2}$$

and

$$\begin{aligned} \mathcal{B}_K &= \rho(K\eta_\rho - 1) + (\mu - \tfrac{1}{2})\eta_\rho + d_\rho\rho(\mu - 1) \\ &\quad - (1 + d_\rho)I_K - \frac{2\rho(2\mu - 1)}{15\mu^{3/2}}. \end{aligned} \tag{3}$$

Lemma 1 (Analytic positivity of the aggregate base derivative). *For*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50},$$

the continuous aggregate reserve defined in (1) satisfies

$$\mathcal{B}_K(\rho, 200) > 0.$$

This assertion uses no numerical or interval certificate.

Proof. We use only the elementary rational bounds

$$3 < \pi < \frac{22}{7}, \quad \frac{78}{25} < \pi, \quad \frac{7}{5} < \sqrt{2}, \quad \frac{265}{153} < \sqrt{3} < \frac{26}{15}. \tag{4}$$

The two square-root bounds in (4) follow immediately by squaring; the displayed rational bounds for π are the classical Archimedean bounds.

Put

$$\mu = 200\rho, \quad b_\rho = \frac{2\pi\rho}{1-\rho}, \quad \bar{R} = 200\rho + \frac{1}{2}, \quad S = \sqrt{\bar{R}^2 + 200b_\rho},$$

and

$$u = \frac{\bar{R}}{\sqrt{200b_\rho}}, \quad P = S - \bar{R}.$$

We first obtain a uniform elementary upper bound for the derivative of the integral term. Since $\rho > 1/8$, one has $\bar{R} < 204\rho$. Therefore

$$u^2 < \frac{204^2 \rho(1-\rho)}{400\pi} \leq \frac{41616}{1600\pi} < \frac{867}{100} < 9.$$

Thus $u < 3$. Moreover

$$\sinh 2 = 2 + \frac{2^3}{3!} + \frac{2^5}{5!} + \cdots > \frac{10}{3} > 3,$$

and hence $\operatorname{arsinh} u < 2$. The identity

$$P = \frac{200b_\rho}{S + \bar{R}}$$

and the inequalities $S > \bar{R} > 200\rho$ give $P < b_\rho/(2\rho)$. Formula (2) now yields

$$I_K(200) = \rho P + \frac{b_\rho}{2} \operatorname{arsinh} u < \frac{3b_\rho}{2}. \quad (5)$$

The final error term in (3) obeys

$$0 < \frac{2\rho(2\mu-1)}{15\mu^{3/2}} < \frac{4\sqrt{\rho}}{15\sqrt{200}} < \frac{2}{105} < \frac{1}{50}. \quad (6)$$

At $K = 200$, formula (3) becomes

$$\begin{aligned} \mathcal{B}_K(\rho, 200) &= 400\rho\eta_\rho - \rho - \frac{\eta_\rho}{2} + d_\rho\rho(200\rho - 1) \\ &\quad - (1 + d_\rho)I_K(200) - \frac{2\rho(400\rho - 1)}{15(200\rho)^{3/2}}. \end{aligned} \quad (7)$$

We first treat $7/51 \leq \rho \leq 1/2$. On this interval

$$\eta_\rho = \omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \quad 0 < d_\rho \leq \frac{1}{3}.$$

The last two bounds in (4) give

$$\frac{13}{120} < \eta_\rho < \frac{1}{9}.$$

Indeed,

$$\frac{\sqrt{3}}{2\pi} > \frac{265}{153} \frac{7}{44} = \frac{1855}{6732} > \frac{11}{40},$$

whereas

$$\frac{\sqrt{3}}{2\pi} < \frac{26}{15} \frac{25}{156} = \frac{5}{18}.$$

The term $d_\rho \rho(200\rho - 1)$ is nonnegative. Combining (5), (6), $2\pi < 44/7$, and (7), we obtain

$$\begin{aligned}\mathcal{B}_K(\rho, 200) &> \rho \left(\frac{127}{3} - \frac{88}{7(1-\rho)} \right) - \frac{17}{225} \\ &\geq \frac{7}{51} \left(\frac{127}{3} - \frac{176}{7} \right) - \frac{17}{225} = \frac{2912}{1275} > 0.\end{aligned}$$

It remains to treat $1/2 \leq \rho \leq 39/50$. Monotonicity of $\arcsin$, together with $\arcsin \rho \geq \rho$, gives

$$\frac{7\rho}{11} < d_\rho < \frac{3}{5}. \quad (8)$$

For the upper bound, note that

$$\frac{39}{50} < \sin \frac{3\pi}{10} = \frac{1 + \sqrt{5}}{4};$$

the strict rational comparison follows from $\sqrt{5} > 53/25$. For the lower bound use $2/\pi > 7/11$.

Since

$$\eta_\rho = G_1(\rho) = \frac{1}{\pi} \int_\rho^1 \arccos t \, dt$$

and $\arccos t \geq \sqrt{2(1-t)}$, we have

$$\eta_\rho \geq \frac{2\sqrt{2}}{3\pi} (1-\rho)^{3/2} > \frac{49}{165} (1-\rho)^{3/2}. \quad (9)$$

Also $0 < \eta_\rho \leq \omega_0 < 1/9$.

Let $[a, c]$ be any of the six intervals in the table below, and let s_c be the displayed rational number. Direct squaring proves $s_c^2 \leq 1 - c$. Hence (9) gives

$$\eta_\rho \geq \eta_c := \frac{49}{165} (1-c)s_c \quad (a \leq \rho \leq c).$$

Equations (5)–(8) then imply the entirely rational lower bound

$$\mathcal{B}_K(\rho, 200) > L(a, c), \quad (10)$$

where

$$L(a, c) := 400a\eta_c - c - \frac{1}{18} + \frac{7a^2}{11}(200a - 1) - \frac{528}{35} \frac{c}{1-c} - \frac{1}{50}.$$

Every entry in the last column is obtained by substitution and integer cross-multiplication:

a	c	s_c	η_c	$L(a, c)$
$\frac{1}{2}$	$\frac{11}{20}$	$\frac{2}{3}$	$\frac{49}{550}$	$\frac{251291}{17325}$
$\frac{11}{20}$	$\frac{3}{5}$	$\frac{5}{8}$	$\frac{49}{660}$	$\frac{70619}{5040}$
$\frac{3}{5}$	$\frac{13}{20}$	$\frac{7}{12}$	$\frac{2401}{39600}$	$\frac{576451}{44100}$
$\frac{7}{10}$	$\frac{7}{10}$	$\frac{6}{11}$	$\frac{147}{3025}$	$\frac{4940821}{435600}$
$\frac{20}{49}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{49}{1320}$	$\frac{2411}{315}$
$\frac{3}{4}$	$\frac{39}{50}$	$\frac{23}{50}$	$\frac{1127}{37500}$	$\frac{11101801}{1386000}$

All six rational lower bounds are strictly positive, and the intervals tile $[1/2, 39/50]$. This proves the assertion on the high branch and completes the proof. $\square$

A purely analytic lower bound for the aggregate reserve at $K = 200$

Abstract

This note proves analytically one base-frequency sign in the aggregate-tail argument. Every definition is reproduced below. The proof uses one concave tangent bound, one Young inequality, elementary rational enclosures, and two exact Bernstein-basis identities. In fact, it proves the uniform, hand-checkable estimate

$$\mathcal{B}(\rho, 200) > 60 \quad \left(\frac{7}{51} \leq \rho \leq \frac{39}{50} \right).$$

No decimal approximation or machine-generated sign enclosure is a premise.

1 The exact assertion

For $0 < \rho < 1$, put

$$G_1(s) = \frac{\sqrt{1-s^2} - s \arccos s}{\pi}, \quad \eta_\rho = G_1\left(\max\{\rho, \tfrac{1}{2}\}\right), \quad d_\rho = \frac{2 \arcsin \rho}{\pi}.$$

At the fixed frequency $K = 200$, define

$$\mu = 200\rho, \quad b = \frac{2\pi\rho}{1-\rho}, \quad c = bK = \frac{400\pi\rho}{1-\rho}, \quad \bar{R} = \mu + \frac{1}{2}, \quad S = \sqrt{\bar{R}^2 + c},$$

and

$$\bar{\mathcal{I}} = \frac{1}{2} \left[\bar{R}S + c \operatorname{arsinh} \frac{\bar{R}}{\sqrt{c}} - \bar{R}^2 \right]. \quad (1)$$

These are exactly the specializations at $K = 200$ of the aggregate-tail quantities in the main proof. The reserve to be estimated is

$$\begin{aligned} \mathcal{B}(\rho, 200) &= (\mu - \tfrac{1}{2})(200\eta_\rho - 1) + \frac{d_\rho}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) \\ &\quad - (1 + d_\rho)\bar{\mathcal{I}} - \frac{8(\mu + \frac{1}{2})}{15\sqrt{\mu}}. \end{aligned} \quad (2)$$

Theorem 1. *For every*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50},$$

one has

$$\mathcal{B}(\rho, 200) > 60 > 0.$$

2 A concave majorant for the integral term

Set

$$H(u) = \frac{1}{2} \left[u\sqrt{1+u^2} + \operatorname{arsinh} u - u^2 \right] = \int_0^u (\sqrt{1+t^2} - t) dt.$$

Scaling (1) by $u = \bar{R}/\sqrt{c}$ gives the exact identity

$$\bar{\mathcal{I}} = cH(u). \quad (3)$$

Lemma 2 (Concave tangent). *For every $u > 0$,*

$$H(u) < \frac{6}{25}u + \frac{12}{25}. \quad (4)$$

Proof. One has

$$H'(u) = \sqrt{1+u^2} - u, \quad H''(u) = \frac{u}{\sqrt{1+u^2}} - 1 < 0.$$

Thus every tangent line is a global upper bound. Choose

$$u_0 = \frac{589}{300}, \quad \sqrt{1+u_0^2} = \frac{661}{300}.$$

Then

$$H'(u_0) = \frac{6}{25}, \quad \operatorname{arsinh} u_0 = \log \left(u_0 + \sqrt{1+u_0^2} \right) = \log \frac{25}{6},$$

and hence

$$H(u_0) - \frac{6}{25}u_0 = \frac{1}{2} \left(\log \frac{25}{6} - \frac{589}{1250} \right). \quad (5)$$

For completeness, the elementary series

$$\log 2 = 2 \sum_{n=0}^{\infty} \frac{1}{(2n+1)3^{2n+1}}$$

gives

$$\log 2 < \frac{2}{3} + \frac{2}{3} \sum_{n=1}^{\infty} 3^{-(2n+1)} = \frac{25}{36}.$$

Also $\log(1+x) < x$ for $x > 0$. Therefore

$$\log \frac{25}{6} = 2 \log 2 + \log \frac{25}{24} < \frac{25}{18} + \frac{1}{24} = \frac{103}{72} < \frac{1789}{1250};$$

the last difference is 29/45000. Substitution in (5) makes the tangent intercept strictly smaller than 12/25, proving (4) by concavity. $\square$

We shall rationalize the remaining square root using the elementary Young inequality

$$\sqrt{c} \leq \frac{c}{2a} + \frac{a}{2} \quad (a > 0), \quad (6)$$

which is equivalent to $(\sqrt{c} - a)^2 \geq 0$. The classical bound

$$\pi < \frac{22}{7} \quad (7)$$

follows, for example, from

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi.$$

Consequently

$$c < c_+(\rho) := \frac{8800}{7} \frac{\rho}{1-\rho}.$$

Combining Lemma 2 with (6) gives, for every $a > 0$,

$$\bar{\mathcal{I}} < Q_a(\rho) := \frac{3\bar{R}c_+(\rho)}{25a} + \frac{3a\bar{R}}{25} + \frac{12c_+(\rho)}{25}. \quad (8)$$

All quantities on the right are rational functions of ρ when a is rational.

3 The coefficient of d_ρ is positive

The replacement of d_ρ by an elementary lower bound requires the sign of its coefficient. We verify that sign before making any such replacement.

Lemma 3. *On $[7/51, 39/50]$,*

$$\frac{1}{2}(\mu - \frac{3}{2})(\mu - \frac{1}{2}) - \bar{\mathcal{I}} > 0. \quad (9)$$

Proof. Write

$$\Phi(\rho) = \bar{R}^2 \frac{1-\rho}{\rho}.$$

Since

$$\frac{d}{d\rho} \log \Phi(\rho) = \frac{200\rho - 400\rho^2 - 1/2}{\rho(1-\rho)\bar{R}},$$

its two critical points are

$$q_\pm = \frac{10 \pm 7\sqrt{2}}{40}.$$

The elementary bounds $7/5 < \sqrt{2} < 3/2$ show

$$q_- < \frac{1}{200} < \frac{7}{51} < q_+ < \frac{41}{80} < \frac{39}{50}.$$

Thus Φ increases and then decreases on the interval in question, so its minimum is attained at one of the two endpoints. From (7),

$$u^2 = \frac{\bar{R}^2(1-\rho)}{400\pi\rho} > \frac{7\bar{R}^2(1-\rho)}{8800\rho}.$$

At the two endpoints the rational lower bounds on the right are

$$\frac{8128201}{2080800}, \quad \frac{685783}{124800},$$

respectively, and

$$\frac{8128201}{2080800} - \frac{225}{64} = \frac{1625777}{4161600} > 0, \quad \frac{685783}{124800} - \frac{225}{64} = \frac{247033}{124800} > 0.$$

Hence $u > 15/8$ throughout the interval.

By (3) and Lemma 2,

$$\frac{\bar{\mathcal{I}}}{\bar{R}^2} = \frac{H(u)}{u^2} < \frac{6}{25u} + \frac{12}{25u^2} \leq \frac{496}{1875}.$$

Moreover $\bar{R} \geq 2851/102 > 27$, and the function

$$\frac{(R-2)(R-1)}{2R^2} = \frac{1}{2} - \frac{3}{2R} + \frac{1}{R^2}$$

is increasing for $R > 4/3$. Therefore

$$\frac{(\mu - \frac{3}{2})(\mu - \frac{1}{2})}{2\bar{R}^2} \geq \frac{25 \cdot 26}{2 \cdot 27^2} = \frac{325}{729}.$$

Finally

$$\frac{325}{729} - \frac{496}{1875} = \frac{82597}{455625} > 0,$$

which proves (9). □

4 Rational lower envelopes

We use only the consequences

$$\frac{1}{\pi} > \frac{7}{22}, \quad \sqrt{2} > \frac{7}{5}, \quad \sqrt{3} > \frac{5}{3}$$

of elementary positive-side squaring and (7).

On $7/51 \leq \rho \leq 1/2$, one has

$$\eta_\rho = G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6} > \frac{13}{132}, \quad d_\rho = \frac{2 \arcsin \rho}{\pi} > \frac{7}{11}\rho.$$

Also $1400/51 \leq \mu \leq 100$, and $(26/5)^2 < 1400/51$, so

$$\frac{8(\mu + 1/2)}{15\sqrt{\mu}} = \frac{8}{15} \left(\sqrt{\mu} + \frac{1}{2\sqrt{\mu}} \right) < \frac{8}{15} \left(10 + \frac{5}{52} \right) = \frac{70}{13}. \quad (10)$$

Define

$$\begin{aligned} \eta_{<} &= \frac{13}{132}, & d_{<} &= \frac{7}{11}\rho, \quad E_{<} = \frac{70}{13}, \\ L_{<}(\rho) &= (\mu - \frac{1}{2})(200\eta_{<} - 1) + \frac{d_{<}}{2}(\mu - \frac{3}{2})(\mu - \frac{1}{2}) - (1 + d_{<})Q_{18}(\rho) - E_{<}. \end{aligned} \quad (11)$$

On $1/2 \leq \rho \leq 39/50$, the integral representation

$$\eta_\rho = \frac{1}{\pi} \int_\rho^1 \arccos t \, dt$$

and $\arccos t \geq \sqrt{2(1-t)}$ give

$$\eta_\rho \geq \frac{2\sqrt{2}}{3\pi}(1-\rho)^{3/2}.$$

Here $1 - \rho \geq 11/50 > (23/50)^2$, and consequently

$$\eta_\rho > \frac{1127}{8250}(1 - \rho) =: \eta_>. \quad (12)$$

The Maclaurin series for $\arcsin \rho$ has positive coefficients, so

$$d_\rho > \frac{7}{11}P_7(\rho) =: d_>, \quad P_7(\rho) = \rho + \frac{\rho^3}{6} + \frac{3\rho^5}{40} + \frac{5\rho^7}{112}. \quad (13)$$

Finally $100 \leq \mu \leq 156 < 169$, whence

$$\frac{8}{15} \left(\sqrt{\mu} + \frac{1}{2\sqrt{\mu}} \right) < \frac{8}{15} \left(13 + \frac{1}{20} \right) = \frac{174}{25} < 7. \quad (14)$$

Define

$$\begin{aligned} E_> &= 7, \\ L_>(\rho) &= (\mu - \tfrac{1}{2})(200\eta_> - 1) + \frac{d_>}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) - (1 + d_>)Q_{67}(\rho) - E_>. \end{aligned} \quad (15)$$

Lemma 3, the lower bounds for d_ρ and η_ρ , the upper bounds for the last term, and (8) now give the logically ordered implications

$$\mathcal{B}(\rho, 200) > L_<(\rho) \quad \left(\frac{7}{51} \leq \rho \leq \frac{1}{2} \right), \quad \mathcal{B}(\rho, 200) > L_>(\rho) \quad \left(\frac{1}{2} \leq \rho \leq \frac{39}{50} \right). \quad (16)$$

The overlap at $\rho = 1/2$ is harmless.

5 Exact Bernstein closure

For $0 \leq x \leq 1$, write

$$B_{k,n}(x) = \binom{n}{k} x^k (1-x)^{n-k}.$$

These functions are nonnegative and sum to one. The conversion from a power polynomial $\sum_{j=0}^n q_j x^j$ to Bernstein coefficients is the finite rational identity

$$\sum_{j=0}^n q_j x^j = \sum_{k=0}^n \left(\sum_{j=0}^k \frac{\binom{k}{j}}{\binom{n}{j}} q_j \right) B_{k,n}(x). \quad (17)$$

Thus every entry below can be checked by additions and cross multiplications of integers.

For the lower interval set

$$x_< = \frac{\rho - 7/51}{1/2 - 7/51} = \frac{102\rho - 14}{37}.$$

Direct substitution of (8) and (11), followed by (17), gives the exact identity

$$(1 - \rho)L_<(\rho) = \frac{1}{D_<} \sum_{k=0}^4 N_{<,k} B_{k,4}(x_<), \quad D_< = 10835145921600, \quad (18)$$

where

k	$N_{<,k}$
0	2977383044192480
1	4628920786869499
2	5747338995812809
3	6988840660610010
4	6621947656848702

The smallest entry is $N_{<,0}$, and

$$N_{<,0} - 60D_{<} = 2327274288896480 > 0.$$

It follows from (18) that

$$(1 - \rho)L_{<}(\rho) > 60. \quad (19)$$

For the upper interval set

$$x_{>} = \frac{\rho - 1/2}{39/50 - 1/2} = \frac{25}{7} \left(\rho - \frac{1}{2} \right).$$

The corresponding exact identity obtained from (8), (15), and (17) is

$$(1 - \rho)L_{>}(\rho) = \frac{1}{D_{>}} \sum_{k=0}^{10} N_{>,k} B_{k,10}(x_{>}), \quad D_{>} = 24873750000000000000, \quad (20)$$

where

k	$N_{>,k}$
0	873011369240936279296875
1	881095006796855712890625
2	884923416376472412109375
3	881704924277462068359375
4	867541422038412498046875
5	837327296894843418528125
6	784608885866591053548975
7	701387295930232136634075
8	577837649792730661053099
9	401902155200474706698937
10	158688669397963044788415

The smallest entry is $N_{>,10}$, and

$$N_{>,10} - 60D_{>} = 9446169397963044788415 > 0.$$

Therefore

$$(1 - \rho)L_{>}(\rho) > 60. \quad (21)$$

Since $0 < 1 - \rho < 1$, both (19) and (21) imply that the corresponding L -function is strictly larger than 60. Combining this conclusion with (16) proves Theorem 1.

Verification boundary. The two Bernstein identities are finite polynomial identities over $\mathbb{Q}$. Formula (17) is a direct hand-checking recipe for every printed integer. No interval evaluation, floating-point comparison, or unprinted case is used in the proof.

A Purely Analytic Aggregate-Tail Theorem

Abstract

We assemble three independently proved analytic base-frequency inequalities with the curvature propagation from the spherical-shell argument. The result replaces the aggregate interval certificate: for $7/51 \leq \rho \leq 39/50$ and $K \geq 200$, the continuous reserve is strictly positive and the discrete aggregate implication proves the Dirichlet Pólya inequality. The inclusion $\rho_c = 1/(1 + 2\pi) > 7/51$ then gives the exact ratio range required in the main proof. No interval enclosure, numerical sign test, or certificate output is used.

1 Definitions and analytic inputs

For $0 \leq s \leq 1$, put

$$G_1(s) = \frac{\sqrt{1-s^2} - s \arccos s}{\pi} = \frac{1}{\pi} \int_s^1 \arccos t \, dt.$$

For $0 < \rho < 1$, define

$$\eta_\rho = G_1\left(\max\{\rho, \tfrac{1}{2}\}\right), \quad d_\rho = \frac{2 \arcsin \rho}{\pi}, \quad b_\rho = \frac{2\pi\rho}{1-\rho}.$$

For $K > 0$, let

$$\begin{aligned} \mu &= \rho K, \quad \bar{R} = \mu + \tfrac{1}{2}, \quad S = \sqrt{\bar{R}^2 + b_\rho K}, \\ \bar{\mathcal{I}}(\rho, K) &= \frac{1}{2} \left[\bar{R}S + b_\rho K \operatorname{arsinh} \frac{\bar{R}}{\sqrt{b_\rho K}} - \bar{R}^2 \right], \end{aligned} \tag{1}$$

and

$$\begin{aligned} \mathcal{B}(\rho, K) &= (\mu - \tfrac{1}{2})(K\eta_\rho - 1) + \frac{d_\rho}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) \\ &\quad - (1 + d_\rho)\bar{\mathcal{I}}(\rho, K) - \frac{8(\mu + \tfrac{1}{2})}{15\sqrt{\mu}}. \end{aligned} \tag{2}$$

Finally, define the curvature floor

$$F(\rho) = 2\rho\eta_\rho + d_\rho\rho^2 - \frac{3(1 + d_\rho)b_\rho}{800}. \tag{3}$$

The three local analytic inputs. The following statements have complete hand proofs in the indicated local notes. They are the only facts imported from those files.

source	imported conclusion on $7/51 \leq \rho \leq 39/50$
<code>B-positive.tex</code>	$\mathcal{B}(\rho, 200) > 60$
<code>BK-positive.tex</code>	$\mathcal{B}_K(\rho, 200) > 0$
<code>F-positive.tex</code>	$F(\rho) > 1549/84150 > 0$

Each source uses only elementary inequalities and finite exact-rational tables. In particular, none of the three conclusions depends on an interval certificate.

The discrete aggregate input from the main analytic argument. Let

$$A_{\rho,1} = \{x \in \mathbb{R}^3 : \rho < |x| < 1\},$$

and let $N_D(A_{\rho,1}, K^2)$ count its Dirichlet eigenvalues strictly below K^2 . When $\mu > 1/2$, write uniquely

$$\mu - \frac{1}{2} = J + \tau, \quad J \in \mathbb{N}_0, \quad 0 \leq \tau < 1,$$

and set

$$R = \begin{cases} J, & \tau = 0, \\ J + 1, & \tau > 0. \end{cases} \quad C = b_\rho K.$$

For $C > 0$, define

$$\mathcal{I}(C, R) = \frac{1}{2} \left[R\sqrt{R^2 + C} + C \operatorname{arsinh} \frac{R}{\sqrt{C}} - R^2 \right]$$

and

$$\mathcal{Q}(\rho, K) = R \lfloor K\eta_\rho \rfloor + d_\rho \frac{J(J+1)}{2} - (1 + d_\rho) \mathcal{I}(C, R) - \frac{8R\tau^{5/2}}{15\sqrt{\mu}}. \quad (4)$$

The already-proved discrete aggregate lemma is the analytic implication

$$\mathcal{Q}(\rho, K) \geq 0 \implies N_D(A_{\rho,1}, K^2) < W(\rho, K) := \frac{2(1 - \rho^3)K^3}{9\pi}. \quad (5)$$

Its proof is the concave-head/convex-tail summation and weighted layer-cake argument; it contains no finite computation. We reproduce below the short continuous-to-discrete comparison needed to invoke it.

Lemma 1 (Continuous reserve implies the discrete reserve). *If $\mu > 3/2$, $K\eta_\rho > 1$, and $\mathcal{B}(\rho, K) \geq 0$, then*

$$\mathcal{Q}(\rho, K) > \mathcal{B}(\rho, K) \geq 0.$$

Proof. Put $y = K\eta_\rho > 1$. The strict floor inequality gives

$$R \lfloor y \rfloor > R(y - 1) \geq (\mu - \tfrac{1}{2})(y - 1).$$

The definition of J, τ, R also gives

$$\frac{d_\rho J(J+1)}{2} > \frac{d_\rho}{2} (\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}).$$

Indeed, with $x = \mu - 1/2 = J + \tau > 1$, the function $x \mapsto x(x - 1)$ is increasing. If $\tau > 0$, then $x(x - 1) < J(J + 1)$; if $\tau = 0$, the difference is $2J > 0$. The function $x \mapsto \sqrt{x^2 + C} - x$ is positive, and

$$\mathcal{I}(C, R) = \int_0^R (\sqrt{x^2 + C} - x) dx.$$

Since $R < \mu + 1/2 = \bar{R}$,

$$\mathcal{I}(C, R) < \mathcal{I}(C, \bar{R}) = \bar{\mathcal{I}}(\rho, K).$$

Finally $R\tau^{5/2} < \bar{R}$. Substitution of these four inequalities in (4) gives exactly $\mathcal{Q}(\rho, K) > \mathcal{B}(\rho, K)$. $\square$

2 Derivative identities and curvature propagation

For fixed ρ , abbreviate $I(K) = \bar{I}(\rho, K)$ and

$$P = S - \bar{R}.$$

Differentiating (1) gives

$$I_K = \rho P + \frac{b_\rho}{2} \operatorname{arsinh} \frac{\bar{R}}{\sqrt{b_\rho K}}, \quad (6)$$

$$I_{KK} = \rho^2 \left(\frac{\bar{R}}{S} - 1 \right) + \frac{\rho b_\rho}{2S} + \frac{b_\rho(2\rho K - 1)}{8KS}. \quad (7)$$

Correspondingly,

$$\mathcal{B}_K = \rho(K\eta_\rho - 1) + (\mu - \tfrac{1}{2})\eta_\rho + d_\rho\rho(\mu - 1) - (1 + d_\rho)I_K - \frac{2\rho(2\mu - 1)}{15\mu^{3/2}}, \quad (8)$$

$$E_{KK} = \frac{\rho^2(2\mu - 3)}{15\mu^{5/2}}, \quad (9)$$

$$\mathcal{B}_{KK} = 2\rho\eta_\rho + d_\rho\rho^2 - (1 + d_\rho)I_{KK} + E_{KK}. \quad (10)$$

These are identities, not estimates.

Lemma 2 (Exact validity conditions on the rational superset). *If*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50}, \quad K \geq 200,$$

then

$$\mu > \frac{3}{2}, \quad K\eta_\rho > 1.$$

Proof. First,

$$\mu = \rho K \geq \frac{1400}{51} > \frac{3}{2}.$$

The function G_1 is decreasing because $G_1'(s) = -\arccos(s)/\pi$. Since $\max\{\rho, 1/2\} \leq 39/50$,

$$\eta_\rho \geq G_1(39/50).$$

The elementary inequality $\arccos t \geq \sqrt{2(1-t)}$ follows on writing $t = \cos \theta$, since $\sqrt{2(1-t)} = 2\sin(\theta/2) \leq \theta$. Combining this with $\pi < 22/7$ gives

$$200\eta_\rho \geq 200G_1(39/50) > \frac{1400\sqrt{2}}{33} \left(\frac{11}{50} \right)^{3/2}.$$

The square of the final positive quantity is

$$\frac{8624}{225} > 1.$$

Thus $200\eta_\rho > 1$, and $K\eta_\rho \geq 200\eta_\rho > 1$. □

Proposition 3 (Analytic curvature propagation). *For*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50}, \quad K \geq 200,$$

one has

$$\mathcal{B}_{KK}(\rho, K) > F(\rho) > \frac{1549}{84150} > 0. \quad (11)$$

Proof. Here

$$S > \bar{R} > \rho K.$$

In (7), the first term is negative. The remaining two terms satisfy

$$\begin{aligned} \frac{\rho b_\rho}{2S} + \frac{b_\rho(2\rho K - 1)}{8KS} &< \frac{\rho b_\rho}{2S} + \frac{2\rho K b_\rho}{8KS} \\ &= \frac{3\rho b_\rho}{4S} < \frac{3b_\rho}{4K} \leq \frac{3b_\rho}{800}. \end{aligned}$$

Therefore

$$I_{KK} < \frac{3b_\rho}{800}.$$

Lemma 2 gives $2\mu - 3 > 0$, so $E_{KK} > 0$. Substitution in (10) yields

$$\mathcal{B}_{KK} > 2\rho\eta_\rho + d_\rho\rho^2 - \frac{3(1 + d_\rho)b_\rho}{800} = F(\rho).$$

The final strict rational bound is precisely the imported conclusion of `F-positive.tex`. $\square$

3 The analytic aggregate-tail theorem

Theorem 4 (Continuous aggregate reserve for all $K \geq 200$). *For every*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50}, \quad K \geq 200,$$

one has the quantitative estimate

$$\mathcal{B}(\rho, K) > 60 + \frac{1549}{168300}(K - 200)^2 > 0. \quad (12)$$

Proof. Fix ρ in the stated interval. The imported local lemmas give

$$\mathcal{B}(\rho, 200) > 60, \quad \mathcal{B}_K(\rho, 200) > 0.$$

Proposition 3 gives

$$\mathcal{B}_{KK}(\rho, s) > \frac{1549}{84150} \quad (s \geq 200).$$

Twice integrating this strict inequality from 200 to K gives

$$\begin{aligned} \mathcal{B}(\rho, K) &= \mathcal{B}(\rho, 200) + (K - 200)\mathcal{B}_K(\rho, 200) + \int_{200}^K (K - s)\mathcal{B}_{KK}(\rho, s) ds \\ &> 60 + \frac{1}{2} \frac{1549}{84150}(K - 200)^2, \end{aligned}$$

including $K = 200$ by the strict imported base value. This is (12). $\square$

Theorem 5 (Purely analytic aggregate tail for spherical shells). *Let*

$$\rho_c = \frac{1}{1 + 2\pi}.$$

For every

$$\rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq 200,$$

the Dirichlet counting function of the unit outer-radius spherical shell satisfies

$$N_D(A_{\rho,1}, K^2) < \frac{2(1 - \rho^3)K^3}{9\pi}.$$

Proof. The exact Archimedean bound $\pi < 22/7$ gives

$$1 + 2\pi < 1 + \frac{44}{7} = \frac{51}{7}, \quad \text{hence} \quad \rho_c > \frac{7}{51}.$$

(Also $\pi > 3$ gives $\rho_c < 1/7 < 39/50$.) Thus the stated ratio interval is contained in the rational superset used in Theorem 4.

On this superset, Lemma 2 verifies $\mu > 3/2$ and $K\eta_\rho > 1$, while Theorem 4 gives $\mathcal{B}(\rho, K) > 0$. Lemma 1 therefore gives $\mathcal{Q}(\rho, K) > 0$. The analytic discrete implication (5) now proves the asserted strict Pólya bound. $\square$

Dependency boundary. The only local-file imports are the three base inequalities listed at the start. The derivative identities, curvature propagation, continuous-to- discrete comparison, rational-domain checks, and two integrations are all displayed in this note. The remaining discrete implication is the previously proved analytic layer-cake lemma from the main manuscript. No aggregate certificate predicate or interval-arithmetic result appears in the dependency chain.

Purely Analytic Closure of the Final Spherical-Shell Residual

Abstract

This note assembles the analytic compact and aggregate estimates after the exact owner subtraction. It proves that the final spherical-shell residual is empty without invoking an interval certificate or an executable truth table. The compact estimate covers a closed superset of the actual sub-200 residual; the aggregate estimate covers the complementary frequency tail.

1 Notation and analytic inputs

Let

$$A_{\rho,1} := \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \quad W(\rho, K) := \frac{2(1 - \rho^3)K^3}{9\pi},$$

and let $N_D(A_{\rho,1}, K^2)$ count Dirichlet eigenvalues strictly below K^2 . Put

$$\rho_c := \frac{1}{1 + 2\pi}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1 - \rho)^2} + 132}.$$

The preceding analytic owner lemmas and the exact set subtraction give

$$\mathcal{D}_{20} = \left\{ (\rho, K) : \rho_c \leq \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho) \right\}. \quad (1)$$

On this interval $U(\rho) = K_0(\rho)$, although the formula for the upper wall will not be needed below.

We use the shell action

$$G_a(z) := \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right) \quad (0 \leq z < a), \quad G_a(z) := 0 \quad (z \geq a),$$

$$A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z),$$

and the strict lattice proxy

$$P(\rho, K) := \sum_{\substack{\nu=\ell+1/2 < K \\ \ell \in \mathbb{N}_0}} 2\nu \# \left\{ n \in \mathbb{N} : n < A_{\rho,K}(\nu) + \frac{1}{4} \right\}. \quad (2)$$

The analytic Bessel-phase reduction proved earlier in the main argument is

$$N_D(A_{\rho,1}, K^2) \leq P(\rho, K). \quad (3)$$

The detailed analytic supplement establishes the following two estimates. They are stated here in exactly the form used in the assembly:

$$P(\rho, K) < W(\rho, K) \quad \left(\rho_c \leq \rho \leq \frac{39}{50}, \quad k_{11}(\rho) \leq K \leq 200 \right), \quad (4)$$

$$N_D(A_{\rho,1}, K^2) < W(\rho, K) \quad \left(\rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq 200 \right). \quad (5)$$

For clarity, the proof of (4) is recalled in the next section. The proof of (5) is the analytic curvature propagation from the three exact base inequalities at $K = 200$.

2 Compact closure

Theorem 1 (Analytic compact theorem). *For*

$$\rho_c \leq \rho \leq \frac{39}{50}, \quad k_{11}(\rho) \leq K \leq 200,$$

one has

$$N_D(A_{\rho,1}, K^2) < W(\rho, K).$$

Proof. The retained-remainder identity in the analytic compact argument gives

$$W - P = \mathcal{B} + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}} - \mathcal{E}_{\text{ang}},$$

with both retained remainders nonnegative. A global primitive minorant then proves the structural implication

$$\mathcal{E}_{\text{ang}}(\rho, K) < \mathcal{H}(\rho, K) \implies P(\rho, K) < W(\rho, K). \quad (6)$$

The paired angular-block lemma, including all common radial-angular jumps, gives

$$\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1 - \rho^2)K^2}{6} - \frac{1}{2}. \quad (7)$$

The scalar positivity theorem proves on the same closed domain that

$$\mathcal{H}(\rho, K) - \frac{(1 - \rho^2)K^2}{6} + \frac{1}{2} > 0. \quad (8)$$

Combining (7) and (8) yields $\mathcal{E}_{\text{ang}} < \mathcal{H}$. Hence (6) gives $P < W$, and (3) completes the proof. $\square$

3 Exact compact/tail ownership

Proposition 2 (Analytic closure of $\mathcal{D}_{20}$). *Define*

$$\mathcal{C}_{21} := \mathcal{D}_{20} \cap \{K \leq 200\}, \quad \mathcal{T}_{21} := \mathcal{D}_{20} \cap \{K > 200\}.$$

Then $\mathcal{C}_{21}$ and $\mathcal{T}_{21}$ are disjoint, exhaust $\mathcal{D}_{20}$, and the strict Pólya inequality holds on both. Consequently the successor residual is empty.

Proof. The two frequency conditions $K \leq 200$ and $K > 200$ form a disjoint partition, so

$$\mathcal{D}_{20} = \mathcal{C}_{21} \dot{\cup} \mathcal{T}_{21}.$$

If $(\rho, K) \in \mathcal{C}_{21}$, then (1) gives

$$\rho_c \leq \rho < \frac{39}{50}, \quad k_{11}(\rho) < K \leq 200.$$

This lies in the closed domain of Theorem 1.

If $(\rho, K) \in \mathcal{T}_{21}$, then

$$\rho_c \leq \rho < \frac{39}{50}, \quad K > 200,$$

so the analytic aggregate estimate (5) applies. Therefore every point of $\mathcal{D}_{20}$ is proved, and

$$\mathcal{D}_{20} \setminus (\mathcal{C}_{21} \cup \mathcal{T}_{21}) = \emptyset.$$

The face $K = 200$ is assigned only to $\mathcal{C}_{21}$; the already owned faces $\rho = 39/50$, $K = k_{11}(\rho)$, and $K = U(\rho)$ remain outside $\mathcal{D}_{20}$. $\square$

Logical dependency. The proof uses finite exact-rational tables only through displayed analytic lemmas. It does not use the former 10,580-box compact certificate, the 1,286-box aggregate certificate, or the 63-state executable subtraction table.